# Do ESG investors care about carbon emissions? Evidence from securitized auto loans*

Christian Kontz

Stanford GSB

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**Abstract**

Securitized auto loans present a clean empirical setting to study the effects of ESG investing on equilibrium asset prices and quantities. I find that the convenience yield of ESG investments increased almost threefold from 0.14% in 2017 to 0.39% p.a. in 2022. The pass-through of this convenience yield to consumer interest rates can be substantial for captive lenders, with implied changes in consumer loan demand ranging from 1.05% to 4.77%. However, I document that the market's focus on firm-level ESG scores, rather than the collateral's $CO_2$ emissions, lowers the cost of capital for high-emission auto ABS by 6 basis points; due to a positive correlation between ESG scores and $CO_2$ emissions. ESG mutual funds allocate more capital to auto ABS from issuers with higher ESG scores, even if those securities finance higher-emission vehicles. These findings highlight that while green premia can have a meaningful impact, they do not necessarily increase the cost of emitting $CO_2$.

JEL classification: G12, G18, G20, G41, Q56

Keywords: Auto ABS, Asset-backed securities, Carbon Emissions, Climate Change, Climate Finance, Corporate Social Responsibility, Environmental, Social, and Governance, Green bonds, Greenium, Green Finance, Sustainable investing, Responsible Investing, SRI, CSR, ESG

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# 1 Introduction

The goal of many environmental, social, and governance (ESG) investors is to raise the cost of emitting $CO_2$ by rewarding "green" assets with lower cost of capital and penalizing "brown" assets with lower cost of capital.[1] The high financing intensity of durable goods, such as automobiles, makes changing their cost of capital a potentially powerful tool to increase the cost of emitting $CO_2$. However, estimating whether investors' non-pecuniary preferences lower the cost of capital of green assets is challenging. A clean measurement must hold risk exposure constant while varying greenness and quantify the true environmental impact of an investment.[2]

This paper sheds light on whether ESG investing lowers the cost of capital for green assets and increases the cost of $CO_2$ emissions by addressing these challenges. I compare the cost of capital across senior tranches of automobile asset-backed securities (auto ABS) using two measures: (i) the issuers' ESG scores and (ii) the collateral pool's $CO_2$ emissions. The unique features of the ABS market allow me to study the effects of ESG investing on equilibrium asset prices and quantities and to identify which dimension of greenness is more influential. Auto ABS are highly standardized debt instruments that securitize pools of auto loans. Only a few parameters distinguish auto ABS deals beyond their collateral pools. Senior tranches of auto ABS are considered safe assets, similar to US Treasurys (Gorton, 2017). I exploit the safe asset nature of senior tranches and use variables derived from loan-level data to control for risk factors across securities. I then test whether the greenness of the security influences its cost of capital and investors' investment decisions. By comparing the impact of firm-level ESG scores and $CO_2$ emissions, I test the often implicit assumption that a green premium increases the cost of emitting $CO_2$. Lastly, I use loan-level data to measure the pass-through elasticity of changes in the cost of capital to consumer interest rates and calculate the corresponding shift in loan demand associated with a green premium.

I leverage loan-level data on the collateral pools of publicly traded consumer auto loan ABS. The data covers 281 auto ABS deals from 2017 to 2022, backed by 17.7 million consumer loans. The securities are issued by 22 issuers, including captive lenders, banks, non-bank finance companies, and vehicle retailers. I estimate the lifetime emissions of each vehicle in a collateral pool by merging the collateral data at the make, model, and year level with $CO_2$ emissions data from the Environmental Protection Agency. The granularity of the loan-level data allows me to control for both predictors and ex-post performance of the collateral pools.

[1] Assets managed by funds using ESG criteria have increased tenfold over the past decade. Asset managers report that the leading ESG criteria are climate change, $CO_2$ emissions, and fossil fuel divestment (USSIF, 2022).

[2] On the paucity of objective measures of greenness, see for example Berg, Koelbel, and Rigobon (2022).

Two unique features of the securitization market allow me to rule out confounders that affect both exposure to risk factors and greenness in my effort to identify the causal effect of ESG investing on asset prices.[3] First, the security design of auto ABS reduces the number of risk factors. Prepayment is the main risk factor for AAA-rated senior tranches of consumer auto loans. Second, consumer and loan characteristics determine prepayment risk rather than the collateral itself. In other words, borrowers with high interest rate loans are more likely to prepay when interest rates fall, regardless of the collateral they finance. This distinction, combined with detailed loan-level data, enables me to disentangle the effects of environmental considerations on equilibrium prices and quantities. The granularity of the loan-level data allows me to control for both ex-ante determinants and ex-post realizations of prepayment risk. Controlling for both predictors and ex-post performance at issuance together with high-dimensional fixed effects removes as much unobserved heterogeneity as possible. This identification strategy allows me to answer the following questions: once an investor can be sure that the loan they extend will be paid back at the agreed time, does the greenness of the deal influence their choice to invest and is this choice reflected in equilibrium prices?

A potential threat to identification is that $CO_2$ emissions or ESG scores might correlate with exposure to risk factors or predictors of cash flow growth. For example, if high-emission pools were less exposed to prepayment risk, then this would provide a rationale for investors to require a lower cost of capital. I therefore verify that neither emissions nor ESG scores predict prepayment at the collateral pool or loan-level. This supports the identifying strategy and increases confidence that differences in the cost of capital are driven by factors unrelated to risk compensation, such as investors' non-pecuniary preferences.

I show that ESG investing successfully lowers the cost of capital of auto ABS for issuers with high firm-level ESG scores. The difference in the cost of capital for high and low ESG assets translate into a convenience yield that investors with non-pecuniary preferences derive from their ESG investment. I estimate that the convenience yield of ESG investments increased almost threefold from 0.14% in 2017 to 0.39% p.a. in 2022. The pass-through of this convenience yield to consumer interest rates is substantial for captive lenders, with implied changes in consumer loan demand ranging from 1.05% to 4.77%.

However, I show that the market's focus on ESG scores instead of the collateral's actual $CO_2$ emission lowers the cost of capital for high-emission auto ABS by 6 bps. I also find that ESG mutual funds, relative to non-ESG, invest more in auto ABS whose issuers have high ESG scores,

[3] I use "ESG investing" as a shorthand for all investment strategies aimed at addressing environmental externalities, including sustainable investing, green investing, socially responsible investing (SRI), ethical investing, and corporate social responsibility (CSR) investing.

even if those securities have a high $CO_2$ emission intensity. These findings are driven by the facts that commonly used firm-level ESG scores are positively correlated with $CO_2$ emissions, do not reflect the variability of emissions across issuers, and do not account for Scope 3 emissions.[4]

I start by documenting the large cross-sectional dispersion of $CO_2$ emissions across auto ABS. For instance, auto ABS issued by Ally Bank finance an average of $55tCO_2$ per vehicle, while those issued by Ford Credit finance an average of $95tCO_2$ per vehicle. This implies that, just as a motorist can choose between high- and low-emission vehicles, investors can choose between auto ABS deals that finance high- or low-emission vehicles. However, the risk-exposure of such an investment is comparable across auto ABS deals. I exploit that auto ABS have large environmental differences but similar risk-characteristics to test whether $CO_2$ emissions impact the cost of capital for auto ABS in equilibrium.

Firm-level ESG and environmental pillar scores do not capture the significant differences in $CO_2$ emissions of auto ABS pools. In fact, both scores are positively correlated with collateral pool emissions when not adjusted for industry differences. Decomposing the variance of ESG scores and $CO_2$ emissions among captive lenders of manufacturers shows that ESG and environmental scores vary significantly less across issuers than their actual emissions.[5] The smaller variance and positive correlation with $CO_2$ emissions make ESG scores poor proxies for the environmental impact of auto ABS.

Using cross-sectional dispersion in collateral $CO_2$ emissions and firm-level ESG scores, I show that estimates of the effects on the cost of capital and conclusions about ESG investors' goal of raising the cost of emitting $CO_2$ depend on accounting for heterogeneity and risk exposure across securities. In a naïve specification that only accounts for market conditions at issuance, auto ABS with emissions below the median have significantly lower issuance spreads of 14.0% to 25.0%. Similarly, issuers with ESG scores above the median see a significant reduction in issuance spreads of 26.8%, indicating alignment between ESG and carbon pricing. However, this naïve approach fails to account for differences in prepayment risk and security design, leading to a biased picture.

When controlling for prepayment risk, I find that *high-emission* collateral pools have a *lower* cost of capital. Moving from the 20th to the 80th percentile of emissions per USD (comparable to moving from Toyota to Ford) reduces issuance spreads by 6 bps, a significant difference compared to the mean issuance spread of 42 bps. Similarly, moving from the 20th to the 80th percentile of

[4]Scope 3 measures indirect emissions that occur in the firm's upstream and downstream supply chain. Scope 3 emissions usually account for more than 70% of a firm's carbon footprint (UN Global Compact).

[5]While it is unclear how and why firm ESG scores of banks should reflect emissions from vehicle loans, one might expect ESG scores of vehicle manufacturers to reflect vehicle emissions. However, I find that this is not the case.

ESG scores reduces issuance spreads by 9 bps, consistent with the positive correlation between ESG scores and emissions. This finding is robust across different measures, tranches, specifications, and estimators. A horse race between emissions and ESG scores shows that ESG scores dominate emissions in explaining the cost of capital of auto ABS. This suggests that investors prioritize ESG scores over actual emissions when evaluating the environmental impact of auto ABS. These findings are robust across different measures, tranches, specifications, and estimators.

The difference in spreads between high and low ESG assets translates into a convenience yield for investors with non-pecuniary preferences. This ESG convenience yield is different from a risk premium; it provides seigniorage to issuers of ESG assets rather than compensation for risk exposure. On average, investors earn an extra 0.28% p.a. in ESG convenience yields, which nearly tripled from 0.14% in 2017 to 0.39% in 2022.[6] This trend aligns with the inflows into ESG funds, which grew by roughly $1.1 trillion between 2017 and 2022 (Van der Beck, 2021).

I next examine mutual fund portfolio data to investigate whether $CO_2$ emissions or ESG scores influence investment in auto ABS. Hong and Kacperczyk (2009) argue that norm-constrained investors, such as ESG mutual funds, are more likely to invest in green securities than unconstrained investors. I test this argument using mutual fund portfolio data, as mutual funds are key investors in the auto ABS market, holding up to 85% of senior tranches. Using a stringent set of fixed effects, I estimate ESG funds' preferences for green assets from multiple auto ABS deals held by both ESG and non-ESG funds during the same reporting period.

I find that ESG funds (i) invest across the full distribution of $CO_2$ emissions and (ii) hold larger portfolio shares in higher-emission auto ABS compared to non-ESG funds. ESG funds allocate approximately 20% less capital to auto ABS deals with emissions below the median than non-ESG funds. These findings are difficult to reconcile with common ESG strategies that usually prescribe outright exclusion or best-in-class investment of brown securities.[7] However, the positive correlation between firm-level ESG scores and $CO_2$ emissions can explain these findings. As expected, ESG funds allocate more capital to auto ABS issued by firms with higher ESG scores than non-ESG funds. However, this reliance on ESG scores tilts their portfolios inadvertently toward higher-emission pools compared with non-ESG funds.

Lastly, I estimate the pass-through of changes in the cost of capital for auto ABS to consumer interest rates and the resulting impact on consumer loan demand. Using an instrumental variable

[6] Similar to my findings, Avramov, Lioui, Liu, and Tarelli (2024) estimate an ESG convenience yield for stocks between 0.37% and 0.66% from 2007 to 2022. Krishnamurthy and Vissing-Jorgensen (2012) document an average convenience yield of 0.73% on US Treasurys.

[7] There is an ongoing debate about which strategy ESG investors should follow: exit (divestment/exclusion) or voice (shareholder activism). Broccardo, Hart, and Zingales (2022) analyze the relative effectiveness of these strategies, while Edmans, Levit, and Schneemeier (2022) examine if exclusion or best-in-class investment is more effective.

approach to isolate exogenous components of the cost of capital, I find that the pass-through elasticity of changes in auto ABS spreads to consumer interest rates is large and positive for captive lenders only. Captive auto lenders jointly optimize lending decisions and vehicle sales and frequently subsidize loans to increase car sales (Benetton, Mayordomo, and Paravisini, 2021). Estimates of the pass-through elasticity imply that a 9 bps decrease in the auto ABS spreads translates into an decrease in consumer interest rates of 19 bps to 30 bps for the average loan by a captive lender. The significant pass-through implies that the benefits of ESG convenience yields extend to consumers as lower borrowing costs. The resulting changes in consumer loan demand are substantial, ranging from 1.05% to 4.77%. For a $33,000 loan, this translates into an increase in loan demand of $347 to $1,574.

I innovate in several ways over the literature. First, this paper is the first to study the effects of environmental externalities, ESG scores, and ESG investing on the pricing and holdings of asset-backed securities. I show that in a market for safe assets, the cost of capital for otherwise identical green assets can significantly differ from brown assets. I exploit a unique setting to compare the impact of ESG scores and $CO_2$ emissions on the cost of capital, highlighting the tension between ESG investing goals and the use of uninformative firm-level ESG scores. I demonstrate that green premia can have a meaningful impact but do not necessarily increase the cost of emitting $CO_2$.

Second, I study the pricing of Scope 3 emissions, which result from product usage and are the largest and most relevant emissions for most firms. Prior studies have focused on Scope 1 and 2 emissions from the production process. For example, Bolton and Kacperczyk (2021) examine the effect of these emissions on stock prices. I find that investors do not consider Scope 3 emissions when pricing auto ABS but focus on firm-level ESG scores, which are positively correlated with financed $CO_2$ emissions.

Third, I measure the pass-through ABS pricing to consumer rates and explore the impact of ESG investing on real quantities like loan demand. By examining how differences in the cost of capital for green and brown assets influence loan demand, I provide insights into the broader economic implications of ESG investing. This analysis helps to understand whether ESG preferences can translate into tangible economic behaviors and outcomes.

Finally, I use *continuous* and *objective* measures of sustainability rather than discrete definitions or proxies like "green" certifications. Other studies on the green premium in debt markets often focus on municipal and corporate bonds, which are either green or not based on the projects they fund (e.g., Baker, Bergstresser, Serafeim, and Wurgler, 2022, Larcker and Watts, 2020). However, green projects can vary greatly in their true environmental impact. I address this by focusing on a single, continuous dimension of environmental relevance.

The paper is organized as follows. The remainder of the introduction discusses the related literature. Section 2 describes the data. Section 3 provides an overview of the auto ABS market and establishes stylized facts about the emissions content of auto ABS and the ESG scores of issuers. Section 4 outlines a green asset pricing model, discusses the identification strategy, and reports estimates of the influence of ESG investors on the cost of capital for auto ABS. Section 5 examines mutual funds' holdings of green and brown auto ABS. Section 6 explores the pass-through of changes in the cost of capital for auto ABS to consumer interest rates and the implied changes in consumer loan demand. Section 7 provides a discussion. Section 8 concludes.

**Related Literature** The rise of ESG investing has spurred extensive research.[8] Theoretical studies suggest that if ESG investors comprise a significant share, green assets will have a lower cost of capital than brown assets. Heinkel, Kraus, and Zechner (2001) model an equilibrium where ESG investor behavior increases the cost of capital for polluting firms and decreases it for green firms. Oehmke and Opp (2024) outline conditions under which ESG investors affect firm behavior, considering social costs and financing constraints. Pástor, Stambaugh, and Taylor (2021) examine how changes in sustainability preferences impact asset prices in equilibrium. Berk and van Binsbergen (forthcomning) analyze the theoretical effects of equity divestment in a single-period mean-variance framework.

Empirical studies of the green premium in debt markets find results which are *prima facie* similar to the results of this study. For example, Pástor, Stambaugh, and Taylor (2022) report that Germany's green twin bonds trade at 5 bps lower yields than their non-green counterparts. Baker et al. (2022) estimate a 6 bps green premium in a sample of over 2,000 U.S. municipal and corporate green bonds, while Zerbib (2019) finds a 2 bps green premium.[9] The results show that a green premium does not necessarily increase the cost of emitting $CO_2$, which is often assumed.

A closely related paper is Hartzmark and Shue (2023). The authors show that redirecting capital from brown to green companies could backfire, as green companies have limited room for improvement, while brown companies worsen under tighter financing. Instead of focusing on companies, I examine auto ABS pricing, which directly finances household vehicle purchases. Adjusting the cost of capital in this market could potentially shift consumer demand from brown to green products.

This paper also contributes to the literature on captive finance and loan securitization. Benmelech, Meisenzahl, and Ramcharan (2017) find that the disruption in ABS markets during the Financial Crisis reduced credit supply and vehicle sales. Klee and Shin (2020) study asymmetric

[8] See Gillan, Koch, and Starks (2021) and Hong and Shore (2023) for excellent reviews.
[9] See also Goss and Roberts (2011), Chava (2014), Tang and Zhang (2020), Flammer (2021), Fatica, Panzica, and Rancan (2021), Seltzer, Starks, and Zhu (2022), Aswani and Rajgopal (2022).

information and signaling in the auto ABS market, documenting that lenders signal private information about higher-quality loans by warehousing these loans longer. Benetton et al. (2021) analyze European auto ABS and find that vertical integration of manufacturing and credit provision allows captive auto lenders to increase cash collected from vehicle sales through credit fire sales. Hankins, Momeni, and Sovich (2022) show that captive finance subsidiaries create a channel for trade policy to affect consumer credit. The authors find that captive lenders passed the Trump administration's metal tariffs through to consumers via higher interest rates. Klee, Morse, and Shin (2023) and Bena, Bian, and Tang (2023) both study electric vehicle (EV) financing terms in securitized loans, with Klee et al. (2023) finding that US EV buyers face lower borrowing costs, while Bena et al. (2023) find that European EV buyers face higher borrowing costs.

# 2 Data

This section describes the loan-level data I use to construct greenness measures and the issuance-level data I use in the empirical tests. The data covers all publicly traded consumer loan[10] auto ABS issued from 2017 to 2022, consisting of approximately 17.7 million unique loans from 281 ABS deals issued by 22 issuers.[11]

**ABS deal data** I collect information about the structure of each deal from prospectuses filed with the SEC, which include details on the deal and its tranches, such as issue date, credit rating, coupon, spreads, issuance amounts, weighted average life (WAL), and book-running banks. I calculate issuance spreads as the difference between the issuance yields and yield curve estimates of Filipović, Pelger, and Ye (2022) by matching the maturity to the WAL of a tranche. Table 1 presents summary statistics for the A-2 tranches of each deal. The average deal size is $1.2 billion, with 30% associated with the A-2 tranche. The average spread is 42 bps with a WAL of one year. Captive lenders issue about 42% of deals, and approximately 28% are sub-prime deals. The average deal finances around 63,031 vehicles. A $100,000 investment finances 219 tons of $CO_2$ over the remaining life of the collateral. The average vehicle emits approximately 58 tons of $CO_2$. The average firm-level issuer ESG scores from Refinitiv and S&P are 0.73 and 0.53, respectively.

**Loan-level data** The loan-level data are from the SEC form ABS-EE. Form ABS-EE is part of the post-financial crisis reporting requirements under Regulation AB, that went into effect on

[10] I exclude vehicle lease and dealer floor plan securitizations from the sample due to their different risk characteristics.

[11] These issuers are: Ally Financial, AmeriCredit, BMW Financial, Capital One Bank, CarMax, Carvana, Exeter Finance, Fifth Third Bank, Ford Credit, GM Financial, Honda Finance, Hyundai Capital, JM Family (WOART), JM Family (WOSAT), Mechanics Bank, Mercedes-Benz Financial Services, Nissan Finance, Santander Bank (DRIVE), Santander Bank (SDART), Toyota Motor Credit, USAA Federal Savings Bank, and Volkswagen Credit.

**Table 1:** Summary Statistics of Issuance-Level Data (A-2 Tranches)

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Total Deal Size ($ m) | 1,242.38 | 348.24 | 1,250.00 | 367.31 | 2,663.82 | 281 |
| Tranche Size ($ m) | 366.71 | 131.99 | 362.00 | 42.40 | 746.94 | 281 |
| Weight. Avg. Life (years) | 0.98 | 0.32 | 1.01 | 0.37 | 3.50 | 281 |
| Spread (bps) | 41.68 | 29.10 | 32.29 | 6.13 | 194.22 | 281 |
| Coupon (%) | 1.91 | 1.30 | 1.86 | 0.14 | 5.81 | 281 |
| Subprime ABS | 0.28 | 0.45 | 0.00 | 0.00 | 1.00 | 281 |
| Captive Lender | 0.44 | 0.50 | 0.00 | 0.00 | 1.00 | 281 |
| Number of Loans Receivable | 63,031 | 21,302 | 62,886 | 15,329 | 136,860 | 281 |
| Avg. Loan-to-Value | 0.92 | 0.04 | 0.92 | 0.80 | 0.98 | 281 |
| Avg. Credit Score | 706.19 | 74.87 | 738.34 | 564.98 | 788.46 | 281 |
| Avg. Interest Rate (%) | 7.64 | 5.87 | 4.46 | 1.38 | 21.35 | 281 |
| Avg. Fraction of Original Loan Outstanding | 0.90 | 0.07 | 0.91 | 0.74 | 1.00 | 281 |
| Avg. Warehousing Time (Months) | 9.54 | 4.38 | 9.19 | 1.33 | 21.06 | 281 |
| Financed tCO2 per $100,000 | 219.14 | 39.96 | 211.08 | 107.10 | 311.78 | 281 |
| Financed tCO2 per Vehicle | 57.78 | 12.11 | 54.50 | 40.54 | 101.25 | 281 |
| Refinitiv ESG Score of Issuer | 0.73 | 0.18 | 0.79 | 0.22 | 0.94 | 243 |
| S&P ESG Score of Issuer | 0.58 | 0.26 | 0.70 | 0.07 | 0.92 | 243 |

*Notes:* Sample contains all publicly traded consumer loan auto ABS from 2017 to 2022. Security-level averages are weighted by outstanding loan balance.

November 23, 2016. This regulation mandates that all prospectuses for public offerings of asset-backed securities must submit loan-level information electronically, with monthly updates on loan pool performance. The data includes information on the originator, borrower, and collateral of each loan. Appendix Table A2 presents the summary statistics of the loan-level data. The average borrower in the sample finances $25,822, at 90% loan-to-value, at a 7.84% interest rate for 67 months. Their credit score is 708 and their monthly payment to income ratio is 0.08. The vehicle the average borrower is financing is worth $27,341.

**Emissions data** Data on $CO_2$ emissions come from the EPA. I match these by make, model, and model year to the loan-level data. Estimates of survival-weighted vehicle miles traveled (SVM) by vehicle type are from the EPA Corporate Average Fuel Economy (CAFE) standard simulator, assuming up to 40 years of useful life. The EPA estimates that trucks and SUVs have higher SVM than sedans and compact cars. Appendix Figure B2 shows SVM curves by vehicle type. The average vehicle in the sample is driven 202,963 miles, with 162,450 miles financed by auto ABS proceeds. Over its lifetime, the vehicle emits 79t$CO_2$, with 63t$CO_2$ financed. Emissions vary significantly among the collateral, which includes fully electric vehicles, compact cars, SUVs,

pickup trucks, and other high-emission vehicles.[12]

**Firm-level ESG scores** I collect firm-level ESG scores of auto ABS issuers[13] from Refinitiv and Standard and Poor's (S&P). Both providers create their scores on the basis of publicly available information and penalize companies with limited reporting. The scores are updated annually and are available for 17 of the 22 originators in the sample.[14]

# 3 Securitized Auto Loans and their $CO_2$ Emissions

This section introduces the market for securitized auto loans, presents key concepts and facts. I demonstrate that, like motorists choosing between high- and low-emission vehicles, investors can choose auto ABS that finance high- or low-emission vehicles. Appendix A.3 shows that commonly used ESG scores are positively correlated with $CO_2$ emissions, do not reflect the variability of emissions across issuers, and do not account for large Scope 3 emissions.

## 3.1 ABCs of Consumer Auto ABS

I focus on homogeneous consumer loan auto ABS that finance vehicle purchases by individual consumers. Auto ABS were among the first consumer ABS to enter the market in the 1980s, and by the end of 2021, they accounted for about 18% of total outstanding auto loans in the US, valued at approximately $220 billion.[15] The auto ABS market is divided into prime and sub-prime deals based on the creditworthiness of the underlying loans, with sub-prime deals commanding higher issuance spreads.

Issuers of consumer auto ABS come from various industries, including vehicle manufacturers and their captive lending companies, specialty retailers, banks, and non-bank finance companies. Table 2 highlights the importance of consumer loan securitization for these industries. On average, companies in the sample securitize approximately 45% of their revenues, 10% of their total assets, or 20% of total unit sales annually. Since consumer loan securitizations are a crucial part

[12] The 10 most popular vehicles in the sample exemplify this heterogeneity. These are, in order, Toyota Camry (sedan, on average 60t of $CO_2$ emissions over full lifetime), Toyota RAV4 (SUV, 73t), Toyota Corolla (sedan, 53t), Nissan Rogue (SUV, 62t), Chevrolet Silverado (truck, 120t), Honda Civic (sedan, 51t), Nissan Altima (sedan, 59t), Honda CR-V (SUV, 65t), Honda Accord (sedan, 62t), and Ford F-150 (truck, 114t).

[13] Technically, a special purpose vehicle (SPV) is the issuer of the auto ABS. The SPVs, however, do not have ESG scores and I therefore use the ESG scores of the sponsoring entity (e.g., Santander Bank or Ford Motor Company) and with a slight abuse of terminology refer to them as the issuer's ESG score.

[14] The results are qualitatively and quantitatively unchanged when using Sustainalytics ESG scores from 2017 to 2019 as Appendix Table A6 and Appendix Table A5 show.

[15] See SIFMA, U.S. ABS issuance and outstanding.

**Table 2:** Average Securitization Intensity per Year by Industry Groups

| Firm-level averages by industry: | Banks | Captive Lenders | Retailers | All Industries |
|---|---|---|---|---|
| Vehicles securitized per year | 278,569 | 231,265 | 261,850 | 248,870 |
| Vehicles securitized as share of units sold | | 0.16 | 0.39 | 0.20[1] |
| Amount securitized as share of revenue | 0.83 | 0.32 | 0.24 | 0.45 |
| Amount securitized as share of assets | 0.11 | 0.05 | 0.30 | 0.10 |

*Notes:* $N$=60 firm-years. Securitization include only consumer loans and exclude lease and floor plan securitizations. Revenue and assets are US vehicle lending segment when available, otherwise for overall US segment (S&P Compustat Segment files from 2016 to 2022). Unit sales of manufacturers from www.goodcarbadcar.net. [1]excludes banks.

of the financial value chain, small changes in financing conditions in the auto ABS market can significantly impact the supply of credit (see also Benmelech et al., 2017).

Compared with corporate and municipal bond markets, the security design of the auto ABS market is highly standardized. Only a few parameters distinguish auto ABS deals from each other besides their collateral pool. All 281 deals in the sample are structured as monthly amortizing with higher seniority tranches receiving repayments first. Appendix Figure B1 shows examples of auto ABS deal structures.[16]

The high levels of standardization in the auto ABS market and the safety of senior tranches make auto ABS highly liquid, as the low bid-ask spreads of $0.04 and immediate market impact estimates of $0.01 show. For comparison, this is a similar level of liquidity and a lower level of market risk than agency mortgage-backed securities that trade in the $200 billion daily volume to-be-announced market (He and Mizrach, 2017).

Prepayment is the main risk for investors in senior tranches of consumer auto ABS since time and risk tranching, high levels of over-collateralization, and other credit enhancements mitigate default risk. Prepayment risk arises from early loan repayments by consumers or borrower defaults leading to vehicle repossession. This risk is measured and priced using a prepayment model. Consumer auto ABS deals also have clean-up call options, allowing issuers to call outstanding notes if the pool balance drops below a certain percentage (typically 5% or 10%). However, these options are irrelevant for most senior tranches as they are paid off before reaching the cutoff.

[16]For a description of the corporate bond underwriting process by investment banks, which is similar to the ABS underwriting process, see Siani (2022).

## 3.2 Stylized Facts about $CO_2$ Emissions from auto ABS

My empirical analysis examines the effects of $CO_2$ emissions from vehicles that serve as collateral for auto ABS on issuance prices and quantities purchased by investors. Auto ABSs need to disclose granular loan-level data that allow me to calculate the financed $CO_2$ emissions for each ABS. By matching the vehicle collateral to emission data from the EPA, I can calculate the amount of financed emissions for each deal.

The amount of expected emissions that the collateral pool of auto ABS $b$ is financing is given by the sum over each vehicle $i$ as:

$$\mathbb{E}\left[\text{Financed } CO_2 \text{ Emissions}\right]_b = \sum_{i \in b} \underbrace{CO_2 \text{ Emissions}_i \times \mathbb{E}\left[\text{Survival-Weighted Miles}\right]_i}_{\text{Expected Emissions}} \times \underbrace{\text{LTV}_i \times \text{Outstanding Balance Share}_i}_{\text{Financing Adjustment}}. \quad (1)$$

The first term on the right-hand side of (1) is the $CO_2$ emissions of vehicle $i$ measured in tons of $CO_2$ per mile driven. The second term is the expected survival-weighted vehicle miles traveled over the lifetime of the vehicle. The product of these terms is the total expected lifetime emissions of a new vehicle.[17] The loan-to-value (LTV) part of the financing adjustment of (1) reflects that not all of the expected $CO_2$ emissions are financed through a loan since many consumers make down-payments at the time of purchase. The financing adjustment also considers that loans have different balances at the time of securitized.

I operationalize estimates of the financed $tCO_2$ emissions from Eq. (1) in two ways: (i) as $tCO_2$ per vehicle and (ii) as $tCO_2$ per \$100,000. Both measures can be justified theoretically: per vehicle measures correspond to environmental impact per quantity of output, while per dollar measures correspond to environmental impact relative to the amount of capital invested. I remain agnostic as to which measure is preferable and report empirical results for both measures throughout.

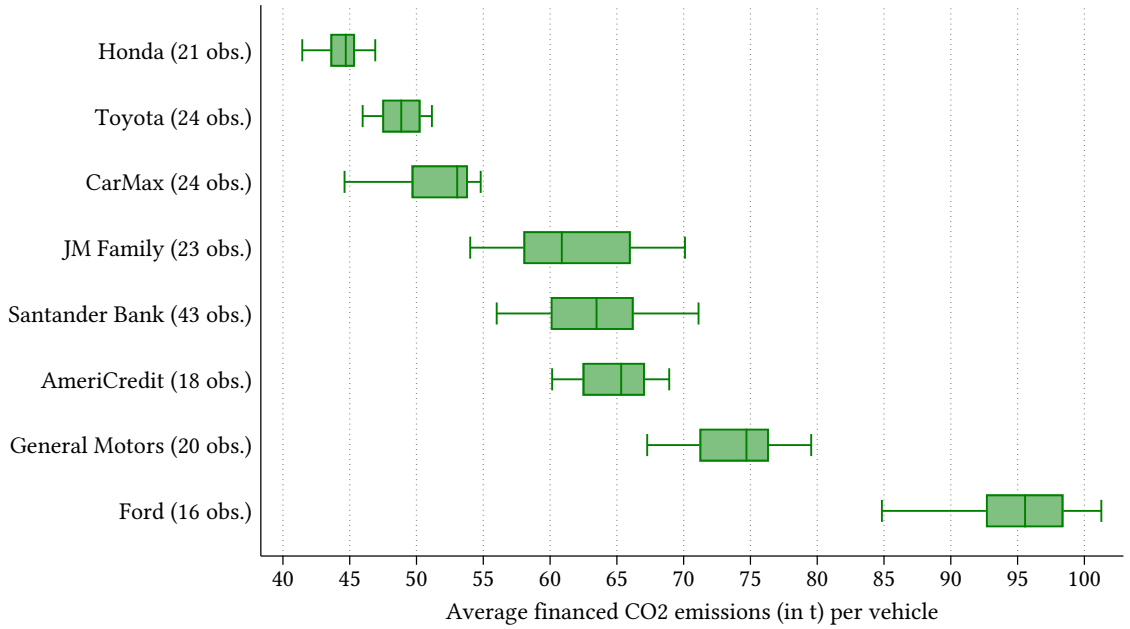
$CO_2$ emissions of ABS deals vary substantially across and within issuers. Figure 1 shows boxplots of average financed $CO_2$ emissions per vehicle across collateral pools for the eight largest issuers.[18] The differences in emissions are largely explained by the vehicle type composition of the collateral pool. Appendix Table A1 shows that a 1 percentage point increase in the truck share raises the average $CO_2$ per vehicle by 1.021 tons.

Figure 1 highlights a key point: just as a motorist can choose between high- and low-emission

[17] I adjust the survival-weighted vehicle miles traveled of used vehicles to reflect the remaining lifetime of the vehicle.

[18] Note that $CO_2$ emissions calculated from Eq. (1) are tailpipe emissions only, excluding production emissions. Appendix Figure B3 shows that including production emissions does not affect the relative rankings.

**Figure 1:** Dispersion of $CO_2$ emissions across all ABS pools of the eight largest issuers




*Notes:* This figure shows boxplots of the average financed $CO_2$ emissions per vehicle across all auto ABS deals for the eight largest issuers by number of deals from 2017 to 2022.

vehicles, investors can choose between auto ABS deals that finance high- or low-emission vehicles. However, the risk-exposure of such an investment is comparable across auto ABS deals. I exploit that auto ABS have large environmental differences but similar risk-characteristics to test whether investors non-pecuniary preferences impact the pricing of auto ABS in equilibrium.

Appendix A.3 shows that the large differences in $CO_2$ intensity are not reflected in the firm-level ESG scores of auto ABS issuers. Commonly used ESG scores are positively correlated with $CO_2$ emissions, do not reflect the variability of emissions across issuers, and do not account for large Scope 3 emissions. The disconnect between ESG scores and $CO_2$ emissions is problematic if investors use ESG scores to screen green from brown auto ABS. In the empirical exercises below, I test whether $CO_2$ emissions of the collateral or issuer's ESG scores have an influence on the cost of capital of auto ABS.

# 4 Issuance Spreads, Cost of Capital, and Convenience Yields

This section develops the main results of the paper that both high-ESG and high-emission auto ABS are associated with lower issuance spreads and therefore lower the cost of capital for auto ABS issuers while earning the holders of those assets a convenience yield. I motivate the empirical specifications through the lens of a simple green asset pricing framework. I then present the

identification strategy and the empirical results.

## 4.1 A simple green asset pricing framework

I assume that the economy is populated by a single investor whose Euler equation is given by

$$\mathbb{E}_t\left[M_{t+1}R^i_{t+1}\right] = \exp(-\beta^i_t\lambda_t) \tag{2}$$

The expression on the left side of the equation is standard. On the right side, I allow the investor to derive a convenience yield $\lambda_t \geq 0$ from holding asset $i$ of $\beta^i_t \in [0,1]$ greenness. Higher values of $\beta^i_t$ correspond to greener assets and earn a convenience yield of $\beta^i_t\lambda_t$. This convenience yield is asset-specific and hence cannot be folded into the SDF. For simplicity, assume that there are only two assets in the economy, a brown $b$ security and a green $g$ security where $\beta^g_t > \beta^b_t$. I assume that $m_t = \log M_t$ and $r^i_t = \log R^i_t$ are conditionally normal. Rewriting the Euler equation (2) using log-normality, one finds

$$\mathbb{E}_t\left[m_{t+1}\right] + \frac{1}{2}\operatorname{Var}_t\left[m_{t+1}\right] + \mathbb{E}_t\left[r^i_{t+1}\right] + \frac{1}{2}\operatorname{Var}_t\left[r^i_{t+1}\right] + \operatorname{Cov}_t\left[m_{t+1}, r^i_{t+1}\right] + \beta^i_t\lambda_t = 0$$

and the following result:

**Lemma 1.** *The expected return in levels on a long position in an asset earned by the investor is decreasing in the convenience yield and in the greenness of the asset:*

$$\mathbb{E}_t\left[r^i_{t+1}\right] - r^f_{t+1} + \sigma^2_{i,t}/2 = -\sigma_{i,m,t} - \beta^i_t\lambda_t \tag{3}$$

Using the Campbell and Shiller (1988) approximation, we can write the dividend yield of an asset with fixed maturity T in this economy as

$$\begin{aligned} dp^i_t &= \sum_{j=0}^{T}\rho^j\mathbb{E}_t\left[r^i_{t+1+j}\right] - \sum_{j=0}^{T}\rho^j\mathbb{E}_t\left[\Delta d^i_{t+1+j}\right] - \kappa\frac{1-\rho^T}{1-\rho} \\ &= -\beta^i_t\lambda_t\frac{1-\rho^T}{1-\rho} + \sum_{j=0}^{T}\rho^j r^f_{t+1+j} - \sum_{j=0}^{T}\rho^j\left(\sigma_{i,m,t,t+j} + \sigma^2_{i,t,t+j}/2\right) - \sum_{j=0}^{T}\rho^j\mathbb{E}_t\left[\Delta d^i_{t+1+j}\right] - \kappa\frac{1-\rho^T}{1-\rho}. \end{aligned} \tag{4}$$

where $\rho = \frac{1}{1-\exp\left(\overline{d-p}\right)}$ and $\kappa = -\log(\rho) - (1-\rho)\log(1/\rho - 1)$. The first term in Eq. (4) shows that a higher non-pecuniary value derived from the greenness of an asset, lowers its dividend yield and thus raises the price of the asset. The differences in yields between two fixed maturity

assets with different levels of greenness but otherwise identical payoffs and risk-characteristics is given by the difference of Eq. (4). I term this difference in yields the green basis.

**Lemma 2.** *The absolute level of the green basis is increasing in final maturity T, the convenience yield, and in the difference of greenness between the two assets:*

$$y_t^g - y_t^b = dp_t^g - dp_t^b = -(\beta_t^g - \beta_t^b)\lambda_t \frac{1-\rho^T}{1-\rho} \tag{5}$$

In the case of a one-period bond, the green basis simplifies to:

$$y_t^g - y_t^b = -(\beta_t^g - \beta_t^b)\lambda_t$$

However, as Eq. (4) shows one, needs to carefully account for potentially differences in risk-exposure and cash-flow growth of green and brown assets in order to infer the green basis. I build an identification strategy in the next section, that exploits the unique features of the auto ABS market to isolate the green basis from risk and cashflow components across auto ABS.

## 4.2 Identification Strategy

Studying whether green assets have a lower cost of capital is challenging. Measuring expected returns is difficult due to equity market volatility[19] and contractual complexities in bonds. Unobserved heterogeneity can create spurious correlations between greenness and risk factors, such as regulatory risk from carbon taxation. This is further complicated by the lack of standardized measures of greenness (Berg et al., 2022), making it hard to quantify environmental impact and differentiate green from non-green securities.

I address these challenges by examining the effects of greenness on the cost of capital in the auto ABS market, using two measures: (i) issuers' ESG scores and (ii) collateral pool $CO_2$ emissions. This market's unique features enable me to identify if green assets have a lower cost of capital and determine which dimension of greenness is more influential. The identification strategy rests on three points.

First, high standardization and the short-term, safe-asset nature of the securities minimize the risk of unobserved heterogeneity affecting the estimates. Additionally, the assets are held in bankruptcy-remote special purpose vehicles, protecting auto ABS holders and collateral pools from sponsor bankruptcy claims. This ensures that issuer-level heterogeneity does not affect risk exposure across auto ABS.

[19] Hanson and Kontz (2024) show that benchmarking of asset managers induces excess co-movement of stocks.

Second, the seniority structure and design of auto ABS deals ensure prepayment is the main risk factor for senior tranches. By allocating cash flows and credit risks specifically across tranches, these deals limit other risks, like credit default, to subordinate tranches. Time tranching prioritizes cash flows to senior tranches, giving them a one-year weighted average life despite typical five-year auto loans. Risk tranching assigns lower default risk to senior tranches and higher default risk to subordinate ones.[20] The analysis focuses on senior tranches rated AAA by at least two agencies. Due to risk tranching, pool-level credit losses would need to reach about 50%, assuming zero recovery value, to affect these tranches, making credit risk an unlikely factor in observed effects.

Third, the granularity of the loan-level data allows me to control for both ex-ante determinants, such as borrower creditworthiness, loan-to-value ratio, and market conditions, as well as ex-post realizations of prepayment risk. Prepayment risk is driven by borrower and loan characteristics, not the greenness of the collateral. Borrowers with high-interest loans tend to prepay when rates fall, regardless of vehicle greenness. I provide evidence that greenness does not predict prepayment, indicating that any observed impact on asset prices and financing costs is due to a preference for green collateral, not greenness as a proxy for risk.

**Empirical Specification** I use the unique features of the auto ABS market to test whether green assets have a lower cost of capital using the following specification:

$$\log\left(\text{Issuance Spread}\right)_{bt} = \alpha \mathbb{1}[\text{Green} > p50]_b + \mathbf{X}_b'\zeta + \gamma_t + \varepsilon_{bt} \tag{6}$$

for bond tranche $b$ issued in year-month $t$. $\mathbb{1}[\text{Green} > p50]_b$ is an indicator variable equal to one if the greenness of the auto ABS deal is above the 50th percentile of all auto ABS deals and zero otherwise. The coefficient of interest is $\alpha$, which reflects the premium investors are willing to pay for a green security. The specification in Eq. (6) is consistent with the literature on the green premium that uses a discrete definition of greenness. In addition, I test the following specification that uses a continuous definition of greenness:

$$\log\left(\text{Issuance Spread}\right)_{bt} = \beta \log\left(\text{Green}\right)_b + \mathbf{X}_b'\zeta + \gamma_t + \varepsilon_{bt} \tag{7}$$

The coefficient of interest is $\beta$, which reflects the elasticity of issuance spreads with respect to greenness. Note that $\gamma_t$ ensures that both $\alpha$ and $\beta$ are identified using the within-month variation

[20] If a borrower defaults, the vehicle is repossessed and sold. For senior tranches, this process is an "involuntary" prepayment. The most junior tranche bears the difference between the outstanding balance and recovery value, with historical recovery values around 60% for prime and 45% for subprime loans (Structured Finance Association).

of greenness across auto ABS deals.

The vector $\mathbf{X}_b$ contains controls for market conditions on the issuance day, tranche characteristics, and collateral pool characteristics. To control for within-month market conditions, $\mathbf{X}_b$ includes estimates of the yield curve for maturity of 6 and 12 months from Filipović et al. (2022), the level of the VIX on the day of issuance, the standard deviation of the VIX in the 30 days before issuance, and the level of the 5-year breakeven inflation rate on the day of issuance.[21] The included tranche characteristics are weighted average life, default attachment point, and the issuance size. The vector $\mathbf{X}_b$ also contains collateral pool controls that are predictors and ex-post realizations of prepayment risk. These are the collateral pool averages of LTV, credit scores, remaining loan balance share, interest rate, warehousing time, realized difference to assumed APS, and the share of loans that are more than 30 days delinquent after 12 months. Collateral pool variables derived from individual loans are weighted by the outstanding loan amount. Collateral pool and tranche characteristic have individual slopes across subprime and prime issues to account for potentially different pricing relevance. All control variables are in logs to achieve a more symmetric and normally distributed data distribution. The specifications further control for subprime issue fixed effects and fixed effects for the assumed absolute prepayment speed (APS). I further interact the APS fixed effects with the weighted average life of the tranche to account for potential "ramp-up" periods in which prepayments increase and then level off to their assumed APS. Standard errors are clustered at the year-month level to account for common variation among spreads caused by market conditions.

**Identifying Assumption** The identifying assumption in both Eq. (6) and Eq. (7) is that the assignment of greenness is uncorrelated with the error term conditional on traditional risk factors: once prepayment risk is accounted for, the assignment of greenness is "as good as random." This identification assumption allows me to answer: does the greenness of a deal influence an investor's choice and is this choice reflected in equilibrium prices? The null hypothesis is that greenness does not affect the cost of capital, implying $\alpha = \beta = 0$. Evidence that $\alpha < 0 < \beta$ would indicate a preference for greener auto ABS deals over brown ones, with investors accepting lower yields.

The assumption would be violated if greenness correlates with risk factors. I test this using ex-post performance variables from monthly auto ABS reports, specifically examining voluntary and involuntary prepayment measures. Appendix Table A3 shows that neither $CO_2$ emissions nor ESG scores predict pool-level performance, with estimates being noisy and close to zero. These results support the assumption, allowing me to treat greenness assignment as effectively

[21] Note that these variables are not subsumed by year-month fixed effects because they vary at daily frequency.

random.

## 4.3 Results

In this subsection, I document the main results: (i) without accounting for risk factors, securities with lower $CO_2$ emissions and those from high-ESG issuers appear to have a significantly lower cost of capital, incorrectly suggesting a alignment between ESG and carbon pricing; (ii) this relationship reverses when accounting for security design and prepayment risk; (iii) the correlation between ESG scores and $CO_2$ emissions confounds the elasticity between emissions and issuance spreads. These findings are robust to various specifications, samples, measures of greenness, and estimators.

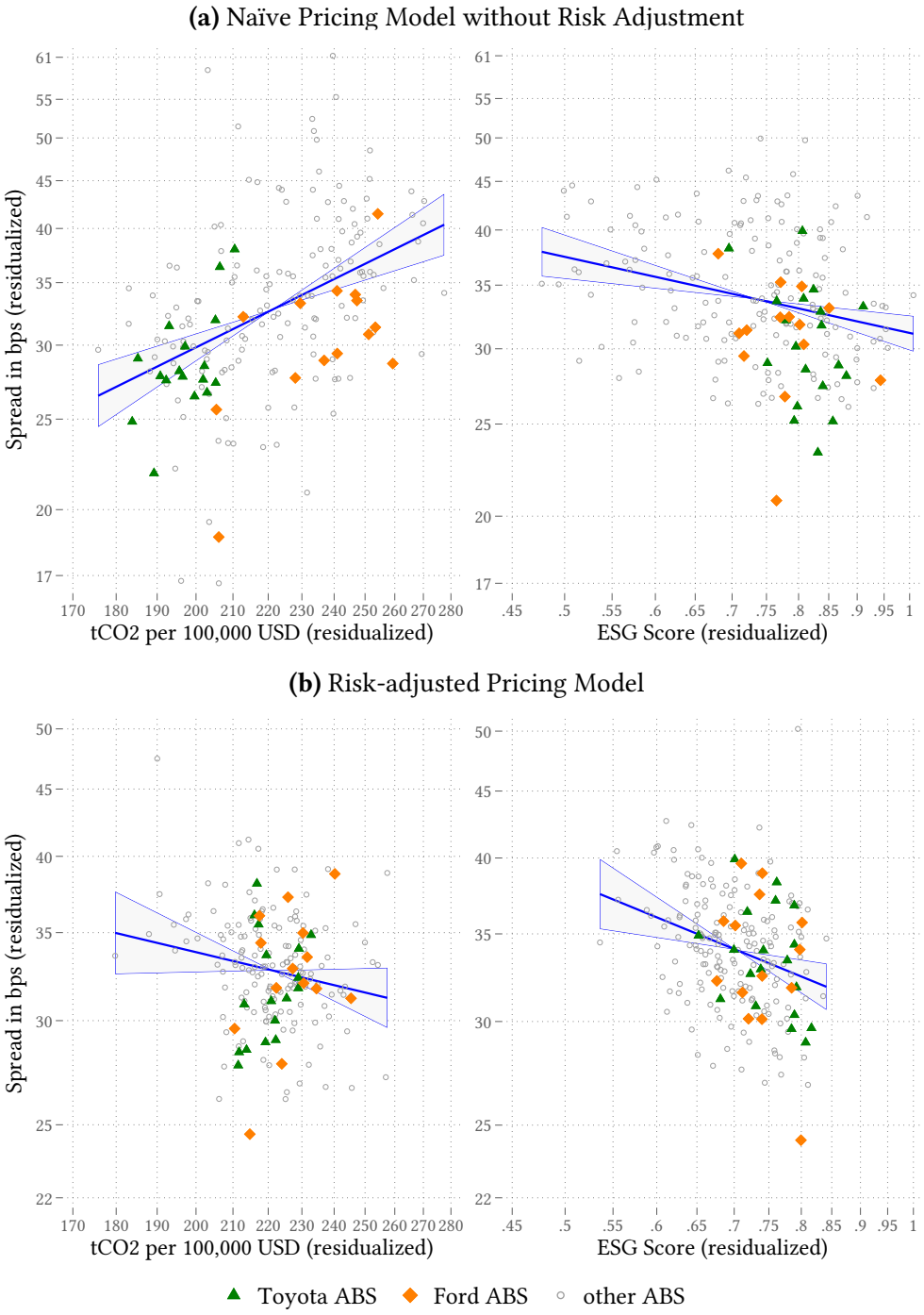
Figure 2a shows that a naive pricing model, controlling only for market conditions, suggests an alignment between ESG and carbon pricing, implying ESG scores can identify environmentally friendly assets and aid climate change mitigation. However, Figure 2b reveals the opposite when using a risk-adjusted model. Controlling for borrower and loan characteristics that determine prepayment risk, which correlates with $CO_2$ emissions, shows that higher emission vehicles are often linked to lower credit scores (Appendix Table A4). Accounting for this, I find that *higher* $CO_2$ emissions are associated with lower issuance spreads, similar to high ESG scores.

Table 3 presents the main results using a risk-adjusted pricing model. Odd-columns controls for ex-ante predictors of prepayment risk, even-columns add controls for ex-post realizations of prepayment risk. Panel A shows estimates of the semi-elasticity of issuance spreads with respect to the high-ESG or low-emission indicator using the pricing model of Eq. (6). Panel B shows estimates of the elasticity of issuance spreads with respect to either ESG scores or $CO_2$ emissions using the pricing model of Eq. (7). Panel C runs a horse race between $CO_2$ emissions and ESG scores, comparing their effects on issuance spreads.

The results in Panel A indicate that high ESG scores lower issuance spreads by between 7.55% and 9.83%.[22] These results are consistent with the hypothesis that investors prefer green assets and are willing to pay a premium for them. However, the results in Panel A also shows that high $CO_2$ emissions lower issuance spreads by 4.39% to 5.61%. This is inconsistent with the hypothesis that investors are concerned with the environmental impact of their investments. Panel B, using the linear pricing model from Eq. (7), reports similar results. Issuance spreads have an elasticity of -0.13 to -0.43 for ESG scores and -0.18 to -0.20 for $CO_2$ emission intensity. Panel C shows the elasticity of issuance spreads while controlling for both ESG scores and $CO_2$ emissions. The elasticity with respect to ESG scores remains significant, but the elasticity for $CO_2$

[22] $-9.83\% \approx 100(\exp\left(-0.103 - 0.0325^2/2\right) - 1)$, see Halvorsen and Palmquist (1980).

**Figure 2:** Residualized Scatter Plots and Line of Best Fit of Pricing Models



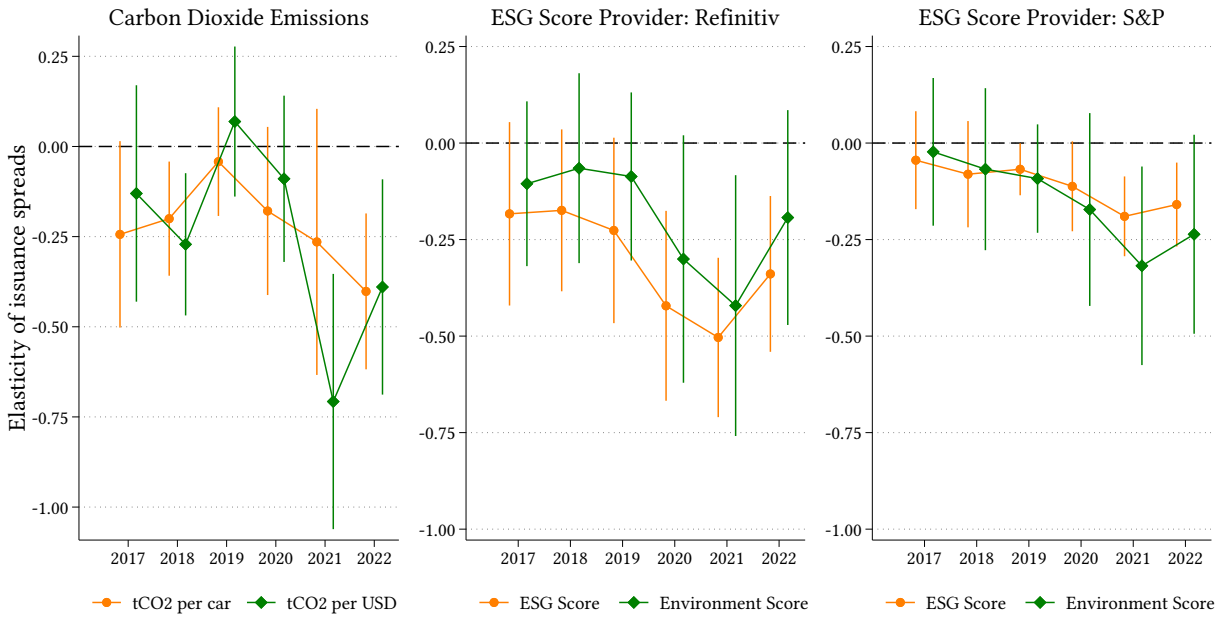

*Notes:* This figure shows pricing models of Eq. (7) in Panel (a) and (b). Regressions in both panels control for issuance-month fixed effects as well as the yield curve, VIX, and inflation expectations which vary within-month. Panel (b) additionally controls for the security design, ex-ante prepayment risk, and ex-post realized prepayment. All variables are in logarithms. Issuance spreads, $tCO_2$ per USD, and ESG scores are trimmed at the 5% and 95% level.

**Table 3:** Both high-ESG and high-emission auto ABS have lower cost of capital

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | | | | Issuance Spread | | | | |
| **Panel A: Semi-Elasticity of Issuance Spreads with respect to High-ESG or Low-Emissions indicator** | | | | | | | | |
| High Refinitiv ESG (score>p50) | -0.103** | -0.078** | | | | | | |
| | (0.033) | (0.029) | | | | | | |
| High S&P ESG (score>p50) | | | -0.096* | -0.085+ | | | | |
| | | | (0.042) | (0.044) | | | | |
| Low Emissions (per USD<p50) | | | | | 0.057* | 0.057* | | |
| | | | | | (0.028) | (0.028) | | |
| Low Emissions (per vehicle<p50) | | | | | | | 0.055* | 0.043 |
| | | | | | | | (0.025) | (0.026) |
| Year-month FE, daily market controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prepayment speed FE, tranche controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-ante prepayment controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-post prepayment controls | | ✓ | | ✓ | | ✓ | | ✓ |
| Adj. $R^2$ | 0.971 | 0.975 | 0.969 | 0.974 | 0.965 | 0.970 | 0.965 | 0.969 |
| Observations | 235 | 235 | 235 | 235 | 276 | 276 | 276 | 276 |
| **Panel B: Elasticity of Issuance Spreads with respect to either ESG score or Carbon Emissions** | | | | | | | | |
| Refinitiv ESG Score | -0.430*** | -0.337** | | | | | | |
| | (0.113) | (0.105) | | | | | | |
| S&P ESG Score | | | -0.131** | -0.135** | | | | |
| | | | (0.047) | (0.047) | | | | |
| Financed tCO2 per USD | | | | | -0.181 | -0.238* | | |
| | | | | | (0.118) | (0.119) | | |
| Financed tCO2 per vehicle | | | | | | | -0.204** | -0.238** |
| | | | | | | | (0.069) | (0.072) |
| Year-month FE, daily market controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prepayment speed FE, tranche controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-ante prepayment controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-post prepayment controls | | ✓ | | ✓ | | ✓ | | ✓ |
| Adj. $R^2$ | 0.972 | 0.976 | 0.970 | 0.975 | 0.965 | 0.970 | 0.966 | 0.971 |
| Observations | 235 | 235 | 235 | 235 | 276 | 276 | 276 | 276 |
| **Panel C: Elasticity of Issuance Spreads with respect to ESG score and Carbon Emissions** | | | | | | | | |
| Refinitiv ESG Score | -0.438*** | -0.341** | -0.409** | -0.291* | | | | |
| | (0.108) | (0.101) | (0.137) | (0.129) | | | | |
| S&P ESG Score | | | | | -0.128** | -0.128** | -0.121* | -0.108* |
| | | | | | (0.046) | (0.046) | (0.047) | (0.046) |
| Financed tCO2 per USD | -0.123 | -0.155 | | | -0.067 | -0.106 | | |
| | (0.121) | (0.122) | | | (0.110) | (0.111) | | |
| Financed tCO2 per vehicle | | | -0.039 | -0.083 | | | -0.127+ | -0.117+ |
| | | | (0.091) | (0.093) | | | (0.066) | (0.068) |
| Year-month FE, daily market controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prepayment speed FE, tranche controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-ante prepayment controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ex-post prepayment controls | | ✓ | | ✓ | | ✓ | | ✓ |
| Adj. $R^2$ | 0.972 | 0.976 | 0.972 | 0.976 | 0.970 | 0.975 | 0.970 | 0.976 |
| Observations | 235 | 235 | 235 | 235 | 235 | 235 | 235 | 235 |

*Notes:* Sample from 2017 to 2022. Panel A shows coefficient estimates of Eq. (6) and Panel B and C show coefficient estimates of Eq. (7). All variables are in logarithms. Standard errors in parentheses are clustered at year-month. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

**Figure 3:** Elasticity of spreads with respect to $CO_2$, ESG, and environment score over time




*Notes:* This figure shows yearly coefficient estimates of the risk-adjusted pricing model of Eq. (7). Standard errors are clustered at the year-month level. Figures shows 90% confidence intervals.

emissions is reduced by up to 50% and often loses statistical significance, suggesting the effect of $CO_2$ emissions is driven by its correlation with ESG scores. Reassuringly, across all panels, estimates accounting for ex-post performance of collateral pools are similar to those controlling only for ex-ante prepayment risk. This supports the assumption that neither ESG scores nor $CO_2$ emissions predict performance.

Figure 4 presents yearly estimates of the elasticity of issuance spreads with respect to $CO_2$ emissions, ESG scores, and environmental pillar scores. These elasticities strengthen over time until 2021, then weaken in 2022. Van der Beck (2021) shows that equity returns from sustainable investing are heavily influenced by price pressure from inflows into sustainable funds. The yearly estimates in Figure 4 closely follow the trend of flows into ESG mutual funds (see Appendix Figure B4). Additionally, Figure 4 indicates that estimates using the environmental pillar scores from both S&P and Refinitiv align closely with the original ESG score estimates, supporting the hypothesis that investors prioritize the environmental impact of their investments.

The results indicate that investors are using ESG or environmental pillar scores to price in their non-pecuniary preferences, while overlooking the actual environmental impact measured by the $CO_2$ intensity of collateral pools. However, it's possible that ESG investors are compensated for high $CO_2$ emissions through higher social and governance scores. To investigate this, I

**Table 4:** Estimates of the ESG Convenience Yield over Time

| | | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | All |
|---|---|---|---|---|---|---|---|---|
| Difference in ESG score: | $\beta_t^g - \beta_t^b$ | 0.29 | 0.18 | 0.20 | 0.15 | 0.43 | 0.31 | 0.32 |
| ESG basis spread in basis points: | $y_t^g - y_t^b$ | -4 | -2 | -2 | -5 | -11 | -12 | -9 |
| ESG Convenience Yield in basis points: | $\lambda_t$ | 14 | 11 | 10 | 34 | 26 | 39 | 28 |
| Average spread of A-2 Tranches in basis points: | | 40 | 38 | 31 | 47 | 22 | 72 | 41 |

*Notes:* This table shows estimates of the ESG convenience yield from 2017 to 2022. Estimates of ESG basis spread are based on Refinitiv ESG scores and yearly elasticity estimates (see Figure 4) using the risk-adjusted pricing model of Eq. (7). Differences in ESG scores evaluated at the 20th and 80th percentiles of ESG score per year.

examined whether deals financing high-emission collateral differ in social and governance scores. The covariate balance test in Appendix Table B1 divides the sample into Green and Brown auto ABS based on $CO_2$ per vehicle and compares ESG, E, S, and G scores from three providers. The table shows that Brown auto ABS generally have slightly higher scores across all dimensions, but most differences are less than 25% of a standard deviation and are not statistically significant. This suggests that ESG investors do not view environmentally worse deals as more desirable in terms of social and governance dimensions.

**Translating yields spreads into convenience yields** The estimated differences in issuance spreads induced by ESG scores (i.e., the green basis) translates into a convenience yield that an investor earns on their ESG investment by rewriting Eq. (5) as

$$\lambda_t = -\frac{y_t^g - y_t^b}{\beta_t^g - \beta_t^b} \tag{8}$$

where $y^g$ and $y^b$ are the yields on green and brown asset and $\beta_t^g$ and $\beta_t^b$ the perceived greenness of the security, respectively. Crucially, this ESG convenience yield is fundamentally different from a risk premium: it generates seigniorage to issuers of ESG assets rather than generating compensation for exposure to a risk factor.

Table 4 shows estimates of the ESG convenience yield based on the ESG scores of Refinitiv and elasticity estimates (see Figure 4) using the risk-adjusted pricing model of Eq. (7). The estimates show that investors receive an average ESG convenience yield of 0.28% p.a.. This convenience yield nearly tripled from 0.14% p.a. in 2017 to 0.39% p.a. in 2022. Similar to my estimates of a 0.28% ESG convenience yield in asset-backed securities, Avramov et al. (2024) estimate an ESG convenience yield for stocks between 0.37% and 0.66%, from 2007 to 2022. These magnitudes compare with other convenience yields, such as the convenience yields on US Treasurys of 0.73% documented by Krishnamurthy and Vissing-Jorgensen (2012).

## 4.4 Robustness

The result that issuers with high ESG scores that finance high-emission auto ABS have a lower cost of capital is robust to using alternative measures, tranches, specifications, and estimators.

**Lower cost of capital for brown and high-ESG auto ABS is robust to different estimators** A potential concern in the main analysis is that the OLS estimators may not accurately control for differences in covariates, falsely attributing differences in issuance spreads to differences in greenness or ESG scores. I address this concern using two alternative estimators.

First, I use the Propensity-Score Matching estimator described by Abadie and Imbens (2016). Appendix Table A13 shows that the matching estimator results for ESG scores are similar to those from the OLS specifications in Panel A of Table 3, whereas the matching estimator results for the low-emission indicator are larger. This likely occurs because the matching estimator selects a sample more similar in terms of covariates than the OLS estimator, suggesting that the main results underestimate the effect of $CO_2$ emissions on issuance spreads. Appendix Figure A2 shows the usual diagnostics, indicating that propensity score matching balances the covariates across auto ABS deals and that propensity scores have large areas of common support.

Second, I use the Double-Lasso estimator from Belloni, Chernozhukov, and Hansen (2014). Appendix Table A14 shows that the Double-Lasso estimator results are qualitatively and quantitatively similar to the main results in Table 3, even when including over 850 potential control variables. The estimator automatically selects relevant control variables for both the outcome and treatment via Lasso estimation. This procedure involves three steps: (1) selecting controls that predict treatment via Lasso, (2) selecting controls that predict the outcome via Lasso, and (3) estimating treatment effects using linear regression while controlling for the union of the selected variables. This method provides inference that is uniformly valid over a large class of models.

**Lower cost of capital for brown auto ABS is robust to alternative measures of greenness** I perform a series of robustness tests using different measures of greenness for each auto ABS deal: (i) expected $tCO_2$ per USD, (ii) expected $tCO_2$ per vehicle, (iii) average MPG of the vehicles in the collateral pool, (iv) average truck share in the collateral pool, and (v) an independently constructed greenness measure by the Kroll Bond Rating Agency (KBRA). [23]

These measures are not perfectly correlated ($0.1 < |\rho| < 0.67$) with the financed $CO_2$ measure from Table 3, providing independent signals about the relative greenness of each auto ABS deal. While measures (ii) to (iv) are highly correlated with the KBRA measure ($|\rho| > 0.75$), the KBRA

[23] (KBRA, 2021, 2022) maps the EPA's vehicle GHG scores (1 to 10, with higher values indicating lower emissions) to 247 auto ABS deals. These scores have been displayed on window labels of new vehicles in the US since 2013.

measure is a good robustness check because it is independently constructed, publicly available, and prominent in the US vehicle market.

Appendix Table A7 shows that the results remain qualitatively unchanged when different measures of relative greenness are used. All specifications indicate that browner auto ABS have a lower cost of capital. Quantitatively, most estimates imply an elasticity of approximately -0.2, which is close to the estimates in the main results of Table 3.

**Lower cost of capital of brown auto ABS holds across the capital structure** The analysis above uses A-2 tranches due to their similar characteristics across different deals: low credit risk, non-binding clean-up call options, and the highest observation count. However, the results are robust to the choice of other AAA-rated tranches. Appendix Table A8 reports results for all AAA-rated tranches, showing qualitatively and quantitatively similar outcomes to the main results in Table 3. The estimated elasticities of issuance spreads with respect to emissions are also close to -0.2 in other tranches, indicating that the lower cost of capital for environmentally worse auto ABS scales through the entire capital structure of these deals.

**Lower cost of capital for brown auto ABS is unrelated to credit quality** A potential concern is that I use both prime and sub-prime auto ABS. Differences in $CO_2$ emissions may be correlated with unobserved characteristics related to loan-quality. For example, subprime borrowers more often buy used vehicles and likely find it harder to refinance their loans or trade in their vehicle than prime borrowers. Appendix Table A9 shows that the result is robust when using prime auto ABS deals only. The estimated elasticities of issuance spreads with respect to emissions are between 0.16 and 0.19 in prime auto ABS, similar to the main result in Table 3. This alleviates potential concerns that the unobserved heterogeneity along credit quality contaminates the original estimates.

**Lower cost of capital of brown auto ABS is not driven by a specific issuer** Given the limited number of 281 deals issued by 22 issuers, there is a concern that a single issuer could be influencing the results. To address this, I re-estimate Eq. (6) and Eq. (7), excluding one issuer at a time. Appendix Figure A1 shows standardized estimates of this leave-out exercise alongside the original estimate, confirming that no single issuer drives the result. The estimated elasticities remain close to the original estimates.

# 5 Auto ABS Holdings of ESG Mutual Funds

In seminal work, Hong and Kacperczyk (2009) argue that norm-constrained investors, such as ESG mutual funds, are more likely to invest in green securities compared to unconstrained investors. To test this, I analyze the portfolio holdings of mutual funds to see if the greenness of auto ABS influences ESG mutual funds' investment decisions.

I document two counterintuitive facts: ESG mutual funds (i) hold positions across the full distribution of $CO_2$ emissions and (ii) invest more in higher-emission deals relative to non-ESG funds. These findings are confounded by the positive correlation between the ESG scores of auto ABS issuers and the $CO_2$ emissions of collateral pools. When accounting for this, it becomes clear that ESG funds invest more in brown auto ABS from issuers with high ESG scores. This further highlights the misalignment between the ESG scores of issuers and the actual greenness of the collateral pools.

**ESG Mutual Funds' Approach to auto ABS** Prospectuses of ESG mutual funds often detail their investment selection approach with regard to asset-backed securities. For example,

"[...] When evaluating securitized debt securities [...], the Adviser generally considers the issuer's ESG rating along with ESG factors related to the underlying pool of assets, such as energy efficiency and environmental impact of the underlying assets"

– ESG Mutual Fund Prospectus I

or

"[...] Potential asset-backed securities are evaluated according to the manager's assessment of material ESG issues for the ABS sectors. The assessment utilizes sector specific metrics across ESG categories, insights from third-party data providers, our analysts' qualitative assessment [...] Environmental assessment involves issues such as carbon emissions, pollution, and renewable energy"

– ESG Mutual Fund Prospectus II

The fact that ESG mutual funds discuss both the "environmental impact of the underlying assets" and the ESG scores of issuers suggests a test to determine which dimension actually influences an ESG fund's investment choice.

## 5.1 Data

I obtain mutual fund holdings from the SEC Form N-PORT from Q3 2019 to Q2 2022. I keep the first observation where a mutual fund reports a position in a senior tranche of an auto ABS. I trim portfolio shares at the 1% level to account for outliers and reporting errors. Appendix Table A10 provides summary statistics of the mutual fund holding data. I observe 266 individual auto ABS deals in the holding data. Mutual funds are among the largest investors in auto ABS, holding an

average of 23% of issuance per deal, with some senior tranches reaching up to 85%. Other notable institutional investors in senior auto ABS include insurance companies, corporate treasuries, and local and state governments.

In the holding data, I identify ESG mutual funds in two ways: (i) by their name using key words[24] such as "sustainable", "ESG", or "climate" and (ii) using a list of "Sustainable Investment Mutual Funds and ETFs" offered by institutional member firms of "The Forum for Sustainable and Responsible Investing".[25] I identify 35 ESG funds (and 787 non-ESG funds) that hold at least one position in an auto ABS tranche over the sample period, with 32 ESG funds holding at least two positions. These ESG funds constitute 25% of the 143 ESG-bond funds identified in the N-PORT filings.

## 5.2 Identification Strategy

To test whether ESG funds tilt their portfolio toward greener auto ABS, I estimate a reduced form asset demand function in the spirit of Koijen and Yogo (2019). I use the following reduced form model for portfolio shares in tranche $t$ of auto ABS deal $b$ by mutual funds $i$ in year-quarter $r$:

$$\log\left(\text{Portfolio Share}\right)_{itrb} = \alpha\left(\text{ESG Fund}_i \times \text{Green}_b\right) + \gamma_i + \gamma_b + \gamma_r + \mathbf{X}_t'\zeta + \varepsilon_{itrb} \tag{9}$$

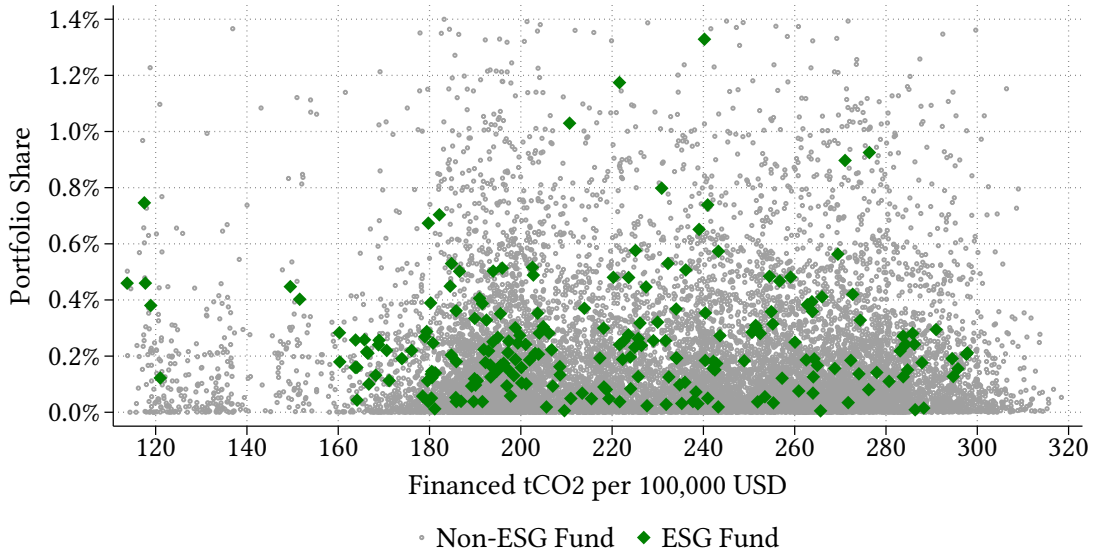
where $\text{Green}_b$ is either a measure of environmental impact such as $tCO_2$ per vehicle, a measure of energy efficiency such as MPG, or the ESG score of the issuer; $\gamma_i$ are fund fixed effects; $\gamma_b$ are auto ABS deal fixed effects; and $\gamma_r$ reporting year-quarter fixed effects. The vector $\mathbf{X}_t$ contains tranche characteristics that are allowed to have different slopes for subprime and prime issues. These characteristics are the weighted average life, size, and annualized yield of the tranche. The coefficient of interest is $\alpha$ which captures the effect of variation in greenness of the collateral on portfolio shares by ESG funds relative to non-ESG funds.

I leverage mutual fund holdings in multiple auto ABSs to determine if ESG funds invest more in environmentally friendly auto ABS compared to non-ESG funds. Using a stringent set of fixed effects, I estimate ESG fund preferences based on the variation in greenness of multiple auto ABS deals held by ESG and non-ESG funds during the same period. The empirical specifications include fixed effects for the collateral pool and fund, thus absorbing the characteristics and preferences of each mutual fund and the specific features of each auto ABS deal. This approach controls for unobserved heterogeneity across collateral pools and funds. The relative difference

[24] The key word list contains the following words: "green", "climate", "esg", "sustainable", "environment", "responsible", "impact", "catholic", "social", "sri", "csr", "community", and "justice".

[25] https://charts.ussif.org/mfpc/

**Figure 4:** Portfolio Shares of Mutual Funds in Auto ABS




*Notes:* This figure shows portfolio shares of ESG and non-ESG mutual funds in auto ABS from 2019-Q3 to 2022-Q2. The x-axis is jittered with normally distributed noise for readability.

in preferences for green assets between ESG and non-ESG funds remains identifiable. Standard errors are clustered at the fund level to account for common variation.

## 5.3 Results

Figure 4 plots mutual fund portfolio shares in auto ABS against financed $CO_2$ emissions per \$100,000. The graph shows that ESG mutual funds hold positions across the full distribution of $CO_2$ emissions. This is surprising since common ESG strategies typically involve either outright exclusions of brown assets or best-in-class investments. Counterintuitively, Figure 4 shows that ESG funds hold similar or higher shares in browner auto ABS.

Table 5 provides estimates of the relationship between greenness and ESG ownership using the specification from Eq. (9). The coefficients in Column (1) of Panel A indicate that the greenest 50% of auto ABS receive 20% less capital from ESG funds compared to non-ESG funds. Columns (2) to (6) present similar estimates using other measures of greenness, all showing positive coefficients of similar magnitude. For example, moving from the 10th to the 90th percentile of average financed $CO_2$ per vehicle (a 2.75 standard deviation difference, equivalent to comparing Honda to Ford auto ABS) results in a 0.45 standard deviation higher portfolio share for ESG funds than for non-ESG funds.

Panel B of Table 5 repeats the reduced form demand estimation of Panel A but controls for the ESG scores of auto ABS issuers. The estimates in columns (1) and (4) show that both the S&P

**Table 5:** Reduced Form Asset Demand System of Mutual Fund for Auto ABS

**Panel A: Measures of Environmental Impact of Investment**

| | (1) Portfolio Share | (2) Portfolio Share | (3) Portfolio Share | (4) Portfolio Share | (5) Portfolio Share | (6) Portfolio Share |
|---|---|---|---|---|---|---|
| ESG Fund=1 × Green (tCO2<p50)=1 | -0.221* | | | | | |
| | (0.091) | | | | | |
| ESG Fund=1 × Financed tCO2 per vehicle | | 0.162* | | | | |
| | | (0.069) | | | | |
| ESG Fund=1 × Financed tCO2 per vehicle | | | 0.162* | | | |
| | | | (0.068) | | | |
| ESG Fund=1 × Avg. MPG ×(−1) | | | | 0.206*** | | |
| | | | | (0.061) | | |
| ESG Fund=1 × Truck Share | | | | | 0.136 | |
| | | | | | (0.138) | |
| ESG Fund=1 × Avg. GHG Rating (KBRA)×(−1) | | | | | | 0.233** |
| | | | | | | (0.078) |
| Tranche controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund FE, Deal FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year-Quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tranche FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Adj. $R^2$ | 0.835 | 0.835 | 0.835 | 0.835 | 0.835 | 0.832 |
| Observations | 11,207 | 11,202 | 11,202 | 11,202 | 11,202 | 10,432 |

**Panel B: ESG Scores versus Environmental Impact of Investment**

| | (1) Portfolio Share | (2) Portfolio Share | (3) Portfolio Share | (4) Portfolio Share | (5) Portfolio Share | (6) Portfolio Share |
|---|---|---|---|---|---|---|
| ESG Fund=1 × S&P ESG Score | 0.171*** | 0.137** | 0.166*** | | | |
| | (0.049) | (0.051) | (0.047) | | | |
| ESG Fund=1 × Refinitiv ESG Score | | | | 0.127** | 0.086+ | 0.121** |
| | | | | (0.045) | (0.048) | (0.045) |
| ESG Fund=1 × Financed tCO2 per car | | 0.079 | | | 0.113 | |
| | | (0.074) | | | (0.079) | |
| ESG Fund=1 × Financed tCO2 per USD | | | 0.042 | | | 0.045 |
| | | | (0.062) | | | (0.068) |
| Fund FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| ABS Deal FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year-Quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tranche FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tranche controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Adj. $R^2$ | 0.836 | 0.836 | 0.836 | 0.836 | 0.836 | 0.836 |
| Observations | 9,987 | 9,987 | 9,987 | 9,987 | 9,987 | 9,987 |

*Notes:* This table shows coefficient estimates of Eq. (9). Sample from 2019-Q3 to 2022-Q2. Coefficients are standardized to unit variances. MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse. Standard errors in parentheses clustered at fund-level (min $K$=639). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

and Refinitiv ESG scores are highly correlated with the differential demand by ESG funds relative to non-ESG funds. Additionally, columns (2), (3), (5), and (6) show that controlling for high ESG scores attenuates the differential demand for high-emission auto ABS by up to 50%.

These results indicate that ESG funds hold more auto ABS from issuers with high ESG scores compared to non-ESG funds. While this is not surprising by itself, the positive correlation between ESG scores and $CO_2$ emissions of the collateral means that ESG funds inadvertently hold higher shares of environmentally less friendly auto ABS than non-ESG funds.[26]

# 6 The real effects of ESG investing

Many ESG investors aim to redirect capital towards greener assets to raise the financing costs for environmentally harmful activities, such as carbon emissions. Higher financing costs should, in turn, reduce the demand for these activities, helping to mitigate climate change. Durable goods, like automobiles, have high financing intensity, so changing their cost of capital could be a powerful way to increase the cost of emitting $CO_2$. However, it is uncertain whether altering the cost of capital in financial markets effectively impacts real quantities, such as consumer loan demand.

The integration of the consumer loan market and financial market pricing of loans via securitizations provides a unique setting to study this question. The impact of ESG investing on real quantities such as consumer loan demand depends on the pass-through of spreads in the ABS market to consumer interest rates and the elasticity of consumer loan demand with respect to these rates. The percentage change in consumer loan demand is given by

$$\partial \log \text{Loan demand} = \frac{\partial \log \text{Loan demand}}{\partial \log \text{Interest rate}} \times \frac{\partial \log \text{Interest rate}}{\Delta \text{ABS spreads}} \times \Delta \text{ABS spreads}. \tag{10}$$

where the first term on the right-hand side is the price elasticity of consumer loan demand with respect to consumer interest rates, the second term is the pass-through elasticity (or semi-elasticity) of financial market spreads to consumer interest rates, and the last term is the change in ABS spreads.

**Changes in ABS spreads** The estimates in Table 4 show that auto ABS that are associated with high ESG scores are issued at 9 bps lower spreads, on average. These estimates, taken at face value, may seem low and their impact on real quantities negligible.[27] However, the effect of changes in

[26] Appendix Table B2 shows the strong correlation of ESG scores and $CO_2$ emissions in the mutual fund holding data.
[27] However, as Table 1 shows average issuance spreads of A-2 tranches are quite small at only 41 bps.

ABS spreads on consumer interest rates depend on the pass-through elasticity of changes in ABS spreads to changes in consumer interest rates. It is not immediately clear whether a green basis spread of 9 bps in senior tranches of auto ABS will result in a 9 bps change in consumer interest rates. It is important to note that the green basis spread represents a decrease in the average cost of funding for the safest tranche in a pool of thousands of loans, rather than the marginal cost decrease of funding for a marginal loan. Moreover, captive auto lenders jointly optimize lending and vehicle sales, which further complicate the pass-through of changes in ABS spreads to consumer interest rates. Consequently, standard pass-through formulas from the industrial organization literature, which typically suggest a 1-for-1 pass-through of changes in marginal cost to consumer prices in competitive markets, do not apply in this context.

**Pass-Through Elasticity** The relationship between ABS spreads and consumer interest rates is inherently endogenous, shaped by equilibrium conditions linking funding markets to the consumer market. To address this endogeneity, I use an instrumental variable (IV) approach to isolate exogenous factors affecting auto loan lenders' funding costs. The instruments include: (i) the ICE BofA US Corporate Index Option-Adjusted Spread, and (ii) the leave-one-out mean of auto ABS spreads issued in the same month, excluding the originator. These instruments leverage common variations in funding markets that correlate with individual lenders' actual funding costs. The IV regressions includes a set of high-dimensional fixed effects, ensuring the pass-through elasticity is identified by loans with the same characteristics made either *across issuers within time* or *across time within issuer*. Appendix A.6 provides further details.

Appendix Table A15 presents the estimates of the pass-through elasticity of auto ABS spreads to consumer interest rates. Several points are noteworthy. First, OLS estimates exhibit an upward bias compared to IV estimates. Second, the estimates for captive lenders are larger than those for non-captive lenders. In IV specifications, the pass-through elasticity is essentially zero for non-captive lenders but large and statistically significant for captive lenders only. Captive auto lenders jointly optimize lending decisions and vehicle sales and frequently subsidize loans to increase car sales (Benetton et al., 2021).[28] Using within-time, across-issuer variation, the estimates of the pass-through elasticity are 0.84% and 1.06%. Using within-issuer, across-time variation, the estimates are 0.65% and 0.80%. These estimates imply that a green basis spread of 9 bps in the auto ABS market translates into an decrease in consumer interest rates of 19 bps to 30 bps for the average loan by a captive lender.

[28] Appendix Figure A3 shows that captive lenders increase their supply of subsidized loans when auto ABS issuance spreads are low. Specifically, a 9 bps decrease in issuance spreads is associated with a 6 percentage point increase in the probability of a loan interest rate being subsidized by a captive lender.

**Table 6:** Implied Changes in Individual Consumer Loan Demand (Captive lenders)

| $\frac{\partial \log \text{Loan demand}}{\partial \log \text{Interest rate}}$ | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| | | -0.18 | | | -0.34 | | | -0.50 | |
| $\frac{\partial \log \text{Interest rate}}{\Delta \text{ ABS spread}}$ | 0.65 | 0.80 | 1.06 | 0.65 | 0.80 | 1.06 | 0.65 | 0.80 | 1.06 |
| $\partial \log$ Loan demand | 1.05% | 1.99% | 2.93% | 1.30% | 2.45% | 3.60% | 1.72% | 3.24% | 4.77% |
| $\Delta$ Loan demand in USD | $347.49 | $656.37 | $965.25 | $427.68 | $807.84 | $1,188 | $566.676 | $1,070.39 | $1,574.1 |

*Notes:* This table reports estimates of the implied change in consumer loan demand: $\partial \log \text{Loan demand} = \frac{\partial \log \text{Loan demand}}{\partial \log \text{Interest rate}} \times \frac{\partial \log \text{Interest rate}}{\Delta \text{ABS spreads}} \times \Delta\text{ABS spreads}$. The average change in ABS spread due to ESG pricing is 9 bps, see Table 4. Intensive margin price elasticity of consumer vehicle credit demand are from Argyle et al. (2020), Lukas (2017), and Attanasio et al. (2008). The average loan amount for captive lenders is approximately $33,000. The pass-through elasticity estimates are in percent per bps.

**Price elasticity** I rely on the extensive literature on the price elasticity of consumer vehicle loan demand with respect to interest rates instead of directly estimating the elasticity. Argyle, Nadauld, and Palmer (2020) report causal estimates for the price elasticity of -0.18, with estimates by FICO subgroup ranging from -0.22 to -0.07. Lukas (2017) estimate a loan price elasticity of -0.34. Attanasio, Koujianou Goldberg, and Kyriazidou (2008) report elasticity estimates ranging from -0.09 to -0.82 but cannot reject the null of zero elasticity. Given the considerable range of estimates for intensive margin price elasticities, I report results for elasticities from -0.18 to -0.5.

**Changes in Consumer Loan Demand** Table 6 shows the implied changes in consumer loan demand for captive lenders associated with the ESG pricing in the auto ABS market. I provide a range of estimates based on the estimates of price elasticity of consumer loan demand and pass-through elasticity. The implied percentages changes in loan demand range from 1.05% to 4.77%. To illustrate, consider a $33,000 loan for a vehicle with a 3.34% interest rate. The results in column (1) imply that a 9 bps decrease in the auto ABS spread would result in a 1.05% increase in equilibrium loan demand, or about $347.5. Column (9) implies a change of about $1,574.

It's important to note that changes in individual consumer loan demand do not directly equate to changes in vehicle demand. The estimated changes in loan demand in Table 6 are best understood as intensive margin changes affecting the loan amount for a given vehicle model purchase. This additional loan demand may be used for upgrades or accessories rather than for higher-priced vehicle models. Depending on the elasticities, the implied changes in individual loan demand could finance a better sound system, set of winter tires, or an upgrade to four-wheel drive. However, manufacturers and captive auto lenders benefit from the cumulative increase in loan demand across all loans in they make, potentially leading to a meaningful increase in product demand and profits for manufacturers with high ESG scores.

# 7 Discussion

Many ESG investors aim to change firm financing conditions by rewarding green firms with lower capital costs and penalizing brown firms with higher costs, thereby internalizing environmental externalities like $CO_2$ emissions. Practitioners consider this mechanism crucial for addressing climate change and promoting sustainability.

I document that investors are successfully lowering the cost of capital for auto ABS of issuers with high firm-level ESG scores. This difference translates into an ESG convenience yield of 0.28% per annum, which investors earn on their ESG investments. This yield is economically meaningful and comparable to other green convenience yields in the literature (e.g., Avramov et al. (2024)). Importantly, this ESG convenience yield differs from a risk premium, as it generates seigniorage for issuers of ESG assets rather than compensation for risk exposure.

The pass-through elasticity of changes in auto ABS spreads to consumer interest rates is large and positive for captive lenders. This significant pass-through implies that the seigniorage benefit extends to consumers in the form of lower borrowing costs, potentially enhancing the affordability and adoption of green products and services.

However, the findings also show that investors are not necessarily investing in the most environmentally-friendly securities but in those whose issuers have higher ESG scores, even if these securities have higher $CO_2$ emission intensities. This raises questions about the effectiveness of ESG investment strategies in addressing environmental externalities. Fund managers with ESG mandates may need to re-evaluate their investment processes to ensure they promote environmentally sustainable investing. ESG funds may also need to seek alternative investments better aligned with their objectives and consider the environmental impact of their investments more closely to meet investor expectations.

These findings suggest a need for greater clarity and transparency in ESG labeling and investment processes. Policymakers may need to provide more guidance to the financial sector on what constitutes environmentally sustainable investing and ensure that ESG labels accurately reflect the environmental impact of investments.

ESG regulation in the United States is still in its infancy. The SEC has issued guidance to ensure that ESG labels accurately reflect the environmental impact of investments, encouraging companies to provide comprehensive and transparent disclosures of their ESG practices and impacts. In Europe, similar efforts are underway with the adoption of the Sustainable Finance Disclosure Regulation (SFDR), which requires comprehensive and transparent disclosure of sustainability risks, impacts, and objectives. Emiris, Harris, and Koulischer (2023) examine the im-

pact of the SFDR on portfolio allocation and ESG fund flows, finding that the regulation increased flows to ESG funds, particularly among environmentally-conscious investors, and that funds with higher initial uncertainty about their sustainability benefited most from the disclosure.

Looking forward, it is essential to consider how investor behavior might evolve as more data on the environmental impact of investments becomes available. Technological advancements can improve data quality and availability, potentially driving a shift towards truly green investments. International standards and collaborations could harmonize ESG criteria and reporting, making informed decisions easier for investors across different markets.

# 8 Conclusion

The goal of many ESG investors is to raise the cost of emitting CO2 by rewarding "green" assets with lower capital costs and penalizing "brown" assets with higher capital costs. This paper shows that ESG investing successfully lowers the cost of capital for auto ABS issuers with high ESG scores. The observed differences in issuance spreads imply that investors derive ESG convenience yields averaging 0.28% per annum, nearly tripling from 0.14% in 2017 to 0.39% per annum in 2022. The estimated pass-through of this green convenience yield to consumer interest rates can be significant for captive lenders, resulting in economically meaningful changes in individual consumer loan demand ranging from 1.05% to 4.77%.

However, the market's focus on firm-level ESG scores, rather than the collateral's $CO_2$ emissions, also lowers the cost of capital for high-emission auto ABS by 6 basis points. This is driven by the fact that commonly used firm-level ESG scores are positively correlated with $CO_2$ emissions, do not reflect the variability of emissions across issuers, and do not account for Scope 3 emissions. Consequently, investors relying on ESG scores to allocate capital may inadvertently subsidize high-emission vehicles. I show that ESG mutual funds allocate more capital to auto ABS from issuers with high ESG scores, even if those securities finance high-emission vehicles.

These findings highlight that while green premia can have a meaningful impact, they do not necessarily increase the cost of emitting $CO_2$. This underscores the need for more accurate and comprehensive project-level ESG metrics that truly reflect environmental impact. Policymakers and investors may need to re-evaluate current ESG strategies to ensure they align with broader environmental goals.

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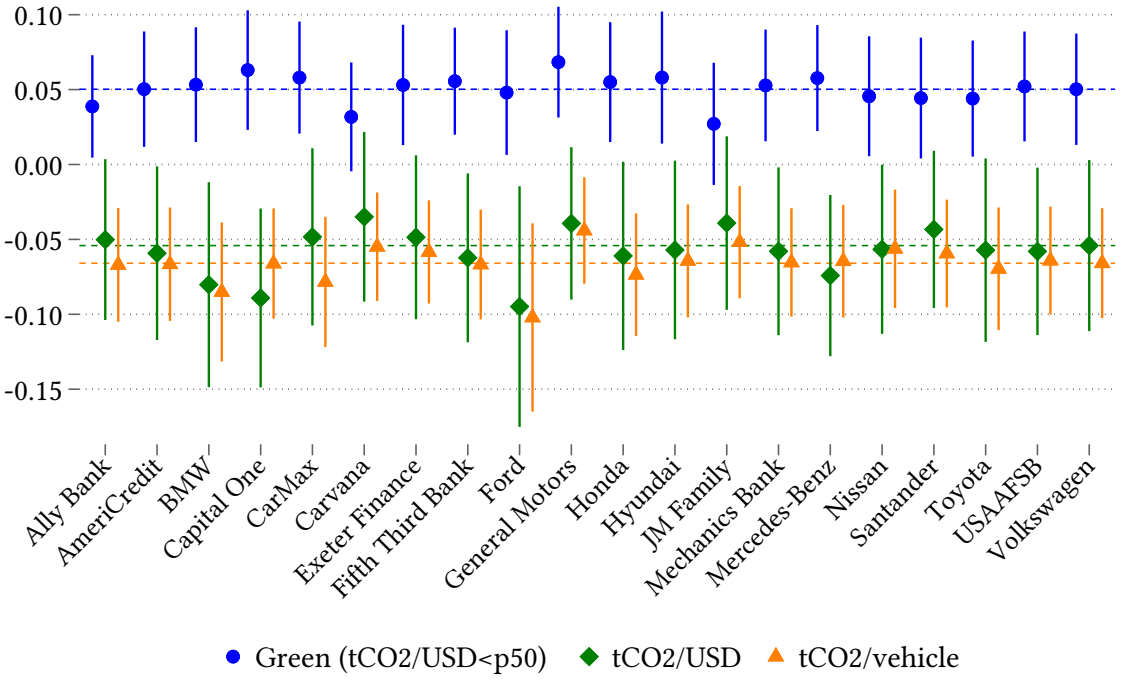
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# A Appendix

## A.1 Appendix Figures




Note: Coefficients are standardized to unit variances. Dashed colored lines give original point estimate.

**Figure A1:** Leave-out estimates of Eq. (6) and Eq. (7) that drop one issuer at a time from the estimation sample.

## A.2 Appendix Tables

**Table A1:** Regression of average $tCO_2$ emissions per vehicle on vehicle types

| | Constant | Truck share | SUV share | Adj. $R^2$ | N | Avg. $tCO_2$/vehicle |
|---|---|---|---|---|---|---|
| $\beta$ | 44.108*** | 1.019*** | 0.210*** | 0.746 | 281 | 70.514 |
| (se) or sd | (1.879) | (0.036) | (0.040) | | | 15.550 |

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

**Table A2:** Summary Statistics Loan-Level Data

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Original Interest Rate | 7.84 | 7.00 | 5.25 | 0.00 | 30.00 | 17,823,551 |
| Original Loan Amount ($) | 25,822.58 | 12,251.91 | 23,650.84 | 518.03 | 248,681.95 | 17,823,552 |
| Original Loan Term (months) | 67.65 | 8.59 | 72.00 | 7.00 | 96.00 | 17,823,552 |
| Credit Score | 708.64 | 101.70 | 719.00 | 250.00 | 900.00 | 17,143,023 |
| Payment-to-Income Share | 0.08 | 0.05 | 0.08 | 0.00 | 0.79 | 17,700,290 |
| Income Verified | 0.09 | 0.29 | 0.00 | 0.00 | 1.00 | 17,823,552 |
| Loan-to-Value | 0.90 | 0.16 | 1.00 | 0.01 | 1.00 | 17,822,211 |
| Outstanding Balance Share | 0.83 | 0.24 | 0.93 | 0.00 | 1.00 | 17,823,548 |
| Vehicle Value Amount ($) | 27,341.46 | 13,177.32 | 24,998.00 | 0.00 | 1,084,455.00 | 17,823,549 |
| Vehicle Age (Years) | 2.74 | 2.56 | 2.00 | 0.00 | 35.00 | 17,823,552 |
| Used Vehicle | 0.48 | 0.50 | 0.00 | 0.00 | 1.00 | 17,823,552 |
| SVM, Financed | 161,660.73 | 40,008.49 | 171,346.10 | 254.15 | 240,728.61 | 17,823,552 |
| SVM, Total | 202,834.40 | 16,986.18 | 207,738.97 | 189,173.82 | 240,728.61 | 17,823,552 |
| tCO2, total Lifetime | 78.28 | 30.61 | 72.45 | 0.00 | 538.75 | 17,823,552 |
| tCO2, remaining Lifetime | 62.12 | 29.51 | 56.48 | 0.00 | 538.75 | 17,823,552 |
| tCO2, financed remaining Lifetime | 46.57 | 27.79 | 44.58 | 0.00 | 538.75 | 17,822,207 |

Notes: Difference to number of receivables in Table 1 are due to not realized upsizing potential of some deals. Credit scores outside the FICO Auto Score range of 250 to 900 are set to missing.

**Table A3:** Greenness and Ex-post Performance of Collateral Pools

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | | Δ Realized APS | | | Realized % loans delinquent 30d+ | | | |
| Financed tCO2 per USD | 0.0921 | | | | 0.0226 | | | |
| | (0.123) | | | | (0.0280) | | | |
| Financed tCO2 per vehicle | | -0.0322 | | | | -0.0305 | | |
| | | (0.139) | | | | (0.0207) | | |
| Refinitiv ESG Score | | | -0.0663 | | | | 0.0795 | |
| | | | (0.159) | | | | (0.0861) | |
| S&P ESG Score | | | | 0.0133 | | | | 0.129 |
| | | | | (0.172) | | | | (0.102) |
| Avg. Credit Score | | | | | -0.534** | -0.559*** | -0.599* | -0.696* |
| | | | | | (0.140) | (0.146) | (0.277) | (0.301) |
| Subprime FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Adj. $R^2$ | 0.0270 | 0.0193 | 0.00902 | 0.00540 | 0.899 | 0.899 | 0.902 | 0.905 |
| Observations | 281 | 281 | 243 | 243 | 281 | 281 | 243 | 243 |

Robust standard errors in parenthesis clustered at originator. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.
Coefficients are standardized to unit variances.

**Table A4:** Covariate Balance Test of *Green* and *Brown* auto ABS Deals

| Variable | (1) Brown (CO2>=p50) Mean/(SE) | (2) Green (CO2<p50) Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Financed tCO2 per USD | 251.861 | 187.059 | 64.802*** |
| | (2.187) | (1.761) | |
| Yield Curve 6m | 0.013 | 0.014 | -0.001 |
| | (0.001) | (0.001) | |
| Yield Curve 12m | 0.014 | 0.016 | -0.002 |
| | (0.001) | (0.001) | |
| VIX | 17.377 | 21.661 | -4.284 |
| | (0.577) | (0.593) | |
| 5 Year Breakeven Inflation | 1.863 | 2.202 | -0.339 |
| | (0.038) | (0.055) | |
| Attachment Point | 0.504 | 0.463 | 0.041* |
| | (0.006) | (0.006) | |
| Weight. Avg. Life | 0.930 | 1.030 | -0.100 |
| | (0.026) | (0.028) | |
| Tranche Size | 336.219 | 397.411 | -61.192 |
| | (10.636) | (11.062) | |
| Loan-to-Value | 0.927 | 0.909 | 0.017 |
| | (0.003) | (0.003) | |
| Mean Credit Score | 673.859 | 738.773 | -64.913*** |
| | (6.297) | (5.018) | |
| Mean Interest Rate | 9.862 | 5.406 | 4.456* |
| | (0.506) | (0.403) | |
| Mean % of outstanding | 0.923 | 0.873 | 0.049 |
| | (0.005) | (0.006) | |
| Warehousing Time (Months) | 8.185 | 10.909 | -2.723 |
| | (0.334) | (0.369) | |
| Number of observations | 141 | 140 | 281 |

Note: Similar to the regression analysis the t-test include APS FE, subprime FE, and year-month FE. * p<0.05, ** p<0.01, *** p<0.001

**Table A5:** Regressions of Sustainalytics ESG scores on $CO_2$ Emissions

| | (1) Sustainalytics ESG score | (2) Sustainalytics Env. score | (3) Sustainalytics ESG score | (4) Sustainalytics Env. score |
|---|---|---|---|---|
| Financed tCO2 per USD | 0.0425 | 0.0174 | | |
| | (0.287) | (0.297) | | |
| Financed tCO2 per vehicle | | | 0.169 | -0.126 |
| | | | (0.145) | (0.145) |
| Issuer Type FE | Yes | Yes | Yes | Yes |
| Adj. $R^2$ | 0.206 | 0.180 | 0.232 | 0.195 |
| Observations | 122 | 122 | 122 | 122 |

Standard errors in parentheses are clustered at issuer-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.
Coefficients are standardized to unit variances.

**Table A6:** Estimates using Sustainalytics ESG scores

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread | (4) Issuance Spread | (5) Issuance Spread | (6) Issuance Spread | (7) Issuance Spread |
|---|---|---|---|---|---|---|---|
| High ESG (score>p50) | -0.0553 | | | | | | |
| | (0.0323) | | | | | | |
| Sustainalytics ESG score | | -0.304** | | -0.389+ | | -0.388+ | |
| | | (0.0951) | | (0.186) | | (0.210) | |
| Sustainalytics Env. score | | | -0.112* | | -0.146 | | -0.156 |
| | | | (0.0518) | | (0.166) | | (0.155) |
| Year-month FE, daily market controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Prepayment speed FE, Tranche FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Other tranche characteristics | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-ante prepayment controls | No | No | No | Yes | Yes | Yes | Yes |
| Ex-post prepayment controls | No | No | No | No | No | Yes | Yes |
| Adj. $R^2$ | 0.941 | 0.943 | 0.940 | 0.947 | 0.945 | 0.948 | 0.946 |
| Observations | 118 | 118 | 118 | 118 | 118 | 118 | 118 |

Standard errors in parentheses are double clustered at issuer and year-month. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

**Table A7:** Elasticity of issuance spreads with respect to different measures of *Greenness*

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread | (4) Issuance Spread | (5) Issuance Spread | (6) Issuance Spread | (7) Issuance Spread | (8) Issuance Spread | (9) Issuance Spread | (10) Issuance Spread |
|---|---|---|---|---|---|---|---|---|---|---|
| Expected tCO2 per USD | -0.199$^{+}$ | -0.245$^{*}$ | | | | | | | | |
| | (0.110) | (0.108) | | | | | | | | |
| Expected tCO2 per vehicle | | | -0.201$^{**}$ | -0.239$^{**}$ | | | | | | |
| | | | (0.0700) | (0.0726) | | | | | | |
| Avg. MPG$\times(-1)$ | | | | | -0.201 | -0.275$^{*}$ | | | | |
| | | | | | (0.127) | (0.130) | | | | |
| Avg. Truck Share | | | | | | | -0.209$^{+}$ | -0.258$^{*}$ | | |
| | | | | | | | (0.106) | (0.121) | | |
| Avg. GHG Rating (KBRA)$\times(-1)$ | | | | | | | | | -0.145 | -0.232$^{+}$ |
| | | | | | | | | | (0.125) | (0.137) |
| Year-month FE, daily market controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Assumed prepayment speed FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Other tranche characteristics | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-ante prepayment controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-post prepayment controls | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes |
| Adj. $R^2$ | 0.966 | 0.970 | 0.966 | 0.971 | 0.965 | 0.970 | 0.965 | 0.970 | 0.961 | 0.968 |
| Observations | 276 | 276 | 276 | 276 | 276 | 276 | 276 | 276 | 243 | 243 |

Standard errors in parentheses clustered at year-month (min K=71). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Average MPG and GHG Rating are multiplied by (-1) such that higher values respond to worse environmental performance.

**Table A8:** Elasticity of issuance spreads with respect to emissions in other senior tranches

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread | (4) Issuance Spread | (5) Issuance Spread | (6) Issuance Spread | (7) Issuance Spread | (8) Issuance Spread |
|---|---|---|---|---|---|---|---|---|
| | A-3 Tranche | | | | A-4 Tranche | | | |
| Financed tCO2 per USD | -0.220$^{**}$ | -0.230$^{**}$ | | | -0.267$^{***}$ | -0.253$^{**}$ | | |
| | (0.0795) | (0.0835) | | | (0.0704) | (0.0742) | | |
| Financed tCO2 per vehicle | | | -0.124$^{*}$ | -0.177$^{**}$ | | | -0.150$^{**}$ | -0.112$^{+}$ |
| | | | (0.0516) | (0.0639) | | | (0.0517) | (0.0628) |
| Year-month FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Market condition controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Assumed prepayment speed FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Other tranche characteristics | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-ante prepayment controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-post prepayment controls | No | Yes | No | Yes | No | Yes | No | Yes |
| Adj. $R^2$ | 0.964 | 0.964 | 0.964 | 0.964 | 0.980 | 0.981 | 0.980 | 0.980 |
| Observations | 272 | 272 | 272 | 272 | 190 | 190 | 190 | 190 |

Standard errors in parentheses clustered at year-month.

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

**Table A9:** Elasticity of issuance spreads with respect to emissions in prime auto ABS

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread | (4) Issuance Spread | (5) Issuance Spread | (6) Issuance Spread | (7) Issuance Spread | (8) Issuance Spread |
|---|---|---|---|---|---|---|---|---|
| Financed tCO2 per USD | -0.139 | -0.167$^{+}$ | | | | | | |
| | (0.0869) | (0.0877) | | | | | | |
| Expected tCO2 per USD | | | -0.166$^{*}$ | -0.184$^{*}$ | | | | |
| | | | (0.0818) | (0.0831) | | | | |
| Financed tCO2 per vehicle | | | | | -0.117$^{*}$ | -0.154$^{**}$ | | |
| | | | | | (0.0515) | (0.0571) | | |
| Expected tCO2 per vehicle | | | | | | | -0.122$^{*}$ | -0.162$^{**}$ |
| | | | | | | | (0.0508) | (0.0562) |
| Year-month FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Market condition controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Assumed prepayment speed FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Other tranche characteristics | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-ante prepayment controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Ex-post prepayment controls | No | Yes | No | Yes | No | Yes | No | Yes |
| Adj. $R^2$ | 0.936 | 0.937 | 0.937 | 0.937 | 0.937 | 0.937 | 0.937 | 0.937 |

Standard errors in parentheses clustered at year-month (min K=70). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

**Table A10:** Summary Statistics of Holding Data

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Portfolio Share | 0.18 | 0.28 | 0.09 | 0.00 | 4.94 | 13,575 |
| Coupon Yield | 2.04 | 1.27 | 1.98 | 0.00 | 8.08 | 13,575 |
| Tranche Size ($m) | 243.86 | 164.98 | 199.34 | 0.00 | 746.94 | 13,575 |
| Weighted Average Life | 2.55 | 1.03 | 2.56 | 0.11 | 5.28 | 13,522 |
| Subprime ABS | 0.47 | 0.50 | 0.00 | 0.00 | 1.00 | 13,575 |
| Financed tCO2 per USD | 232.33 | 38.70 | 237.42 | 118.20 | 311.78 | 13,575 |
| Financed tCO2 per car | 59.71 | 11.62 | 60.05 | 40.54 | 101.27 | 13,575 |

## A.3 ESG scores of issuers and their relationship to $CO_2$ emissions

ESG scores measure a firm's environmental, social, and governance performance. Advocates of ESG investing claim that these scores help investors identify firms with low environmental impact and invest in their securities. The goal of many ESG investors is to alter firm financing conditions by rewarding green firms with lower capital costs and penalizing brown firms with higher costs, thereby internalizing environmental externalities like $CO_2$ emissions. The success of ESG investing depends on investors' ability to accurately identify green firms, projects, and securities.

Appendix Table A11 provides summary statistics for ESG scores of issuers at the time of issuance, as well as measures of the environmental impact of auto ABS collateral pools. The table includes standard summary statistics and decomposes the standard deviation in ESG scores into between ($\overline{sdi}$) and within ($sdit - \overline{sd}_i + \overline{sd}$) components. The between component measures the cross-sectional variation in ESG scores across the 22 issuers, while the within component measures the time series variation in ESG scores for individual issuers. Ratios of between to within standard deviations are reported, with a ratio greater than 1 indicating that cross-sectional variation in ESG scores is larger than individual time series variation.

While it is unclear how the ESG scores of diversified banks and finance companies reflect the emissions of vehicle loans, one might expect the environmental scores of vehicle manufacturers to reflect vehicle emissions. However, this is not the case. Decomposing the standard deviation of ESG scores and $CO_2$ emissions among captive lenders of manufacturers shows that ESG and environmental scores vary considerably less across issuers than their actual emissions footprint.

ESG scores of captive lenders exhibit low variation, while $CO_2$ emission intensity does not. Panel A in Table A11 shows that the standard deviations of ESG scores (0.06 to 0.07) among captive lenders are quite low, with the coefficient of variation for the Refinitiv ESG score being only 8%. Additionally, standard deviation ratios (1.6 to 1.9) indicate that ESG scores for captive lenders vary almost as much within individual time series as they do across issuers.

In contrast, Panel B of Table A11 shows that the standard deviations of $CO_2$ emissions vary 2.6 to 4.9 times more across issuers than within issuers. The coefficient of variation for financed $CO_2$ emissions ranges from 15% to 29%, a fact also illustrated in Figure 1.

Panel C of Table A11 presents the correlations between the environmental impact of auto ABS and the ESG scores of the issuers. While the ESG scores of S&P and Refinitiv are positively correlated with each other, they are also positively correlated with $CO_2$ emissions from collateral pools, even for the environmental pillar scores.

This positive correlation between ESG and environmental pillar scores and pool-level emis-

**Table A11:** ESG and $CO_2$ Summary Statistics of Collateral Pools

| | Mean | SD | $\frac{\text{Between SD}}{\text{Within SD}}$ | Median | Min | Max | N |
|---|---|---|---|---|---|---|---|
| **Panel A: ESG ratings of issuers at time of issuance:** | | | | | | | |
| Refinitiv ESG score of issuer | 0.73 | 0.18 | 4.99 | 0.79 | 0.22 | 0.94 | 243 |
| - Captive Lenders | 0.79 | 0.06 | 1.87 | 0.79 | 0.62 | 0.94 | 123 |
| - Other Lenders | 0.68 | 0.23 | 6.67 | 0.78 | 0.22 | 0.90 | 120 |
| Refinitiv environmental score of issuer | 0.69 | 0.31 | 5.95 | 0.85 | 0.00 | 0.97 | 243 |
| - Captive Lenders | 0.85 | 0.07 | 1.70 | 0.86 | 0.67 | 0.98 | 123 |
| - Other Lenders | 0.66 | 0.36 | 5.96 | 0.53 | 0.00 | 0.92 | 120 |
| S&P ESG score of issuer | 0.58 | 0.26 | 3.43 | 0.70 | 0.07 | 0.92 | 243 |
| - Captive Lenders | 0.63 | 0.17 | 1.63 | 0.67 | 0.27 | 0.86 | 123 |
| - Other Lenders | 0.53 | 0.32 | 9.27 | 0.75 | 0.07 | 0.92 | 120 |
| S&P ESG environmental score of issuer | 0.61 | 0.31 | 4.42 | 0.76 | 0.00 | 0.98 | 243 |
| - Captive Lenders | 0.70 | 0.16 | 1.71 | 0.76 | 0.34 | 0.98 | 123 |
| - Other Lenders | 0.53 | 0.39 | 9.82 | 0.79 | 0.00 | 0.95 | 120 |
| **Panel B: Measures of environmental impact of collateral pools:** | | | | | | | |
| Financed tCO2 per vehicle | 58.01 | 12.76 | 3.39 | 54.49 | 40.54 | 101.27 | 281 |
| - Captive Lenders | 58.58 | 17.24 | 4.98 | 50.48 | 41.33 | 101.27 | 123 |
| - Other Lenders | 57.75 | 7.55 | 2.00 | 58.46 | 40.73 | 71.11 | 156 |
| Financed tCO2 per $100,000 | 219.58 | 40.08 | 1.77 | 211.15 | 107.10 | 311.78 | 281 |
| - Captive Lenders | 197.62 | 29.99 | 2.59 | 199.49 | 107.10 | 274.82 | 123 |
| - Other Lenders | 237.15 | 38.58 | 1.05 | 241.35 | 150.28 | 311.78 | 156 |
| Expected tCO2 per $100,000 | 292.83 | 51.42 | 1.61 | 296.31 | 161.51 | 456.16 | 281 |
| Expected tCO2 per vehicle | 70.51 | 15.55 | 4.15 | 67.61 | 42.94 | 125.73 | 281 |
| Average Miles-per-Gallon per vehicle | 24.25 | 2.49 | 2.80 | 23.88 | 18.71 | 32.66 | 281 |
| Average EPA GHG rating per vehicle | 5.59 | 0.54 | 2.31 | 5.68 | 4.13 | 6.68 | 247 |

**Panel C: Correlation between ESG scores of issuers and environmental impact of collateral pools:**

| | Refinitiv ESG score | Refinitiv Env. score | S&P ESG score | S&P Env. score | Fin. tCO2 per vehicle | Fin. tCO2 per USD | Avg. MPG |
|---|---|---|---|---|---|---|---|
| Refinitiv ESG score of issuer | 1.00 | | | | | | |
| Refinitiv environmental score of issuer | 0.87 | 1.00 | | | | | |
| S&P ESG of issuer | 0.73 | 0.69 | 1.00 | | | | |
| S&P environmental score of issuer | 0.77 | 0.72 | 0.96 | 1.00 | | | |
| Financed tCO2 per vehicle | 0.41 | 0.27 | 0.31 | 0.28 | 1.00 | | |
| Financed tCO2 per USD | 0.11 | 0.03 | 0.20 | 0.12 | 0.58 | 1.00 | |
| Average MPG $\times(-1)$ | 0.23 | 0.07 | 0.08 | 0.07 | 0.80 | 0.44 | 1.00 |

Notes: Financed and expected tons of $CO_2$ of collateral pools calculated using Eq. (1). EPA GHG rating per vehicle as calculated by KBRA (2022). Spearman rank correlation among N=243 observations for which ESG scores of issuers are available. MPG is multiplied by (-1) such that higher values correspond to worse environmental performance.

sions is not entirely surprising. ESG scores are composite firm-level measures of the overall societal impact of a firm, whereas the $CO_2$ measures of auto ABS pools are project-specific measures of environmental impact. I also provide results using the firm-level environmental pillar score, which should be more comparable to the $CO_2$ intensity of auto ABS.

One potential explanation for the positive correlations is that the project-specific environmental impact of the collateral pool does not reflect the firm-level environmental impact of the issuer's overall business. Another explanation could be that the correlation matrix in Panel C of Table A11 fails to account for industry differences in ESG scores.

Table A12 explores the positive correlation between ESG scores and the environmental impact of auto ABS by regressing ESG and environmental pillar scores on firm-level and pool-level

**Table A12:** Regressions of ESG scores on $CO_2$ Emissions

| | (1) Refinitiv ESG score | (2) S&P ESG score | (3) Refinitiv Env. score | (4) S&P Env. score | (5) Refinitiv ESG score | (6) S&P ESG score | (7) Refinitiv Env. score | (8) S&P Env. score |
|---|---|---|---|---|---|---|---|---|
| **Panel A: Auto ABS emissions intensity:** | | | | | | | | |
| Financed tCO2 per USD | 0.0661 | 0.0896 | 0.00435 | 0.00780 | | | | |
| | (0.103) | (0.150) | (0.0738) | (0.125) | | | | |
| Financed tCO2 per vehicle | | | | | 0.152 | -0.0570 | 0.0943 | -0.0550 |
| | | | | | (0.0994) | (0.185) | (0.0880) | (0.162) |
| Industry fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Adj. $R^2$ | 0.583 | 0.447 | 0.587 | 0.473 | 0.604 | 0.446 | 0.596 | 0.476 |
| Observations | 243 | 243 | 243 | 243 | 243 | 243 | 243 | 243 |
| **Panel B: Firm-year emissions intensity from Trucost:** | | | | | | | | |
| Scope 1+2/Revenue | 0.132 | 0.213 | 0.138 | 0.184 | | | | |
| | (0.158) | (0.238) | (0.160) | (0.219) | | | | |
| Scope 3 Downstream/Revenue | | | | | 0.0911 | 0.0895 | 0.148 | 0.0822 |
| | | | | | (0.118) | (0.204) | (0.113) | (0.180) |
| Industry fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Adj. $R^2$ | 0.487 | 0.317 | 0.501 | 0.372 | 0.539 | 0.307 | 0.524 | 0.366 |
| Observations | 99 | 99 | 99 | 99 | 83 | 83 | 83 | 83 |

Standard errors are clustered at issuer-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001 Coefficients are standardized to unit variances.

measures of carbon intensity while accounting for industry fixed effects. The results in Panel A show that the positive correlation between ESG scores and the environmental impact of auto ABS is attenuated but not driven by industry differences in ESG scores. Qualitatively similar results are obtained when using sub-scores on which the environmental pillar score is based, such as emissions scores.

Panel B shows that even at the firm level, ESG and environmental pillar scores are positively correlated with $CO_2$ emissions. Scope 1 and 2 emissions are positively correlated with ESG scores, even when controlling for industry fixed effects. Columns (1) to (4) show this for scope 1 and 2 emissions over revenue, while Columns (5) to (8) show this for scope 3 emissions over revenue. Additionally, the collateral pool $CO_2$ per USD is directly related to the scope 3 downstream emission intensity of vehicle manufacturers.[29]

This implies that the observed positive correlation between ESG and environmental pillar scores and $CO_2$ emissions of collateral pools is not just a data artifact but indicates that ESG and environmental pillar scores are uninformative about $CO_2$ emissions among auto ABS issuers, even at the firm level.

[29]Unreported regressions confirm the strong relationship between the collateral pool level $CO_2$ per USD and the scope 3 downstream emission intensity of vehicle manufacturers as reported by Trucost.

**Table A13:** Estimates using Propensity Score Matching

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread |
|---|---|---|---|
| Green (tCO2<p50) | 0.236*** | | |
| | (0.0616) | | |
| Top-ESG (Refinitiv Score>p50) | | -0.136* | |
| | | (0.0590) | |
| Top-ESG (S&P Score>p50) | | | -0.128* |
| | | | (0.0563) |
| Time, Subprime, APS FE | Yes | Yes | Yes |
| Observations | 84 | 174 | 198 |
| Treated | 50 | 93 | 77 |
| Control | 34 | 81 | 121 |
| # Nearest Neighbors | 2 | 2 | 2 |

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

## A.4 Matching Estimator

Appendix table A13 shows that one obtains qualitatively and quantitatively similar results to the main results when using a propensity score matching estimator. The "treated" (i.e., either low $CO_2$ emissions or high ESG score) and "untreated" auto ABS are matched to their k=2 nearest neighbors.

Appendix Figure A2 provides diagnostics on the validity of the propensity matching. Panel (a) to (c) show that the propensity scores of $CO_2$ , Refinitiv's ESG, as well as S&P's ESG score have sufficient overlap to estimate a matching model. The red bars in panels (a) to (c) show propensity score restrictions imposed on the region of overlap when estimating the models of Appendix table A13. Panel (d) shows the standardized difference[30] between treated and untreated units. The standardized difference are all within or close the rule of thumb bands ($\pm 0.25$) depicted as red bars. This implies that a regression model adjusting for covariates would not be sensitive to model specification.

[30] $\Delta x = \frac{\bar{x}_{\text{treated}} - \bar{x}_{\text{untreated}}}{\sqrt{S^2_{\text{treated}} - S^2_{\text{untreated}}}}$

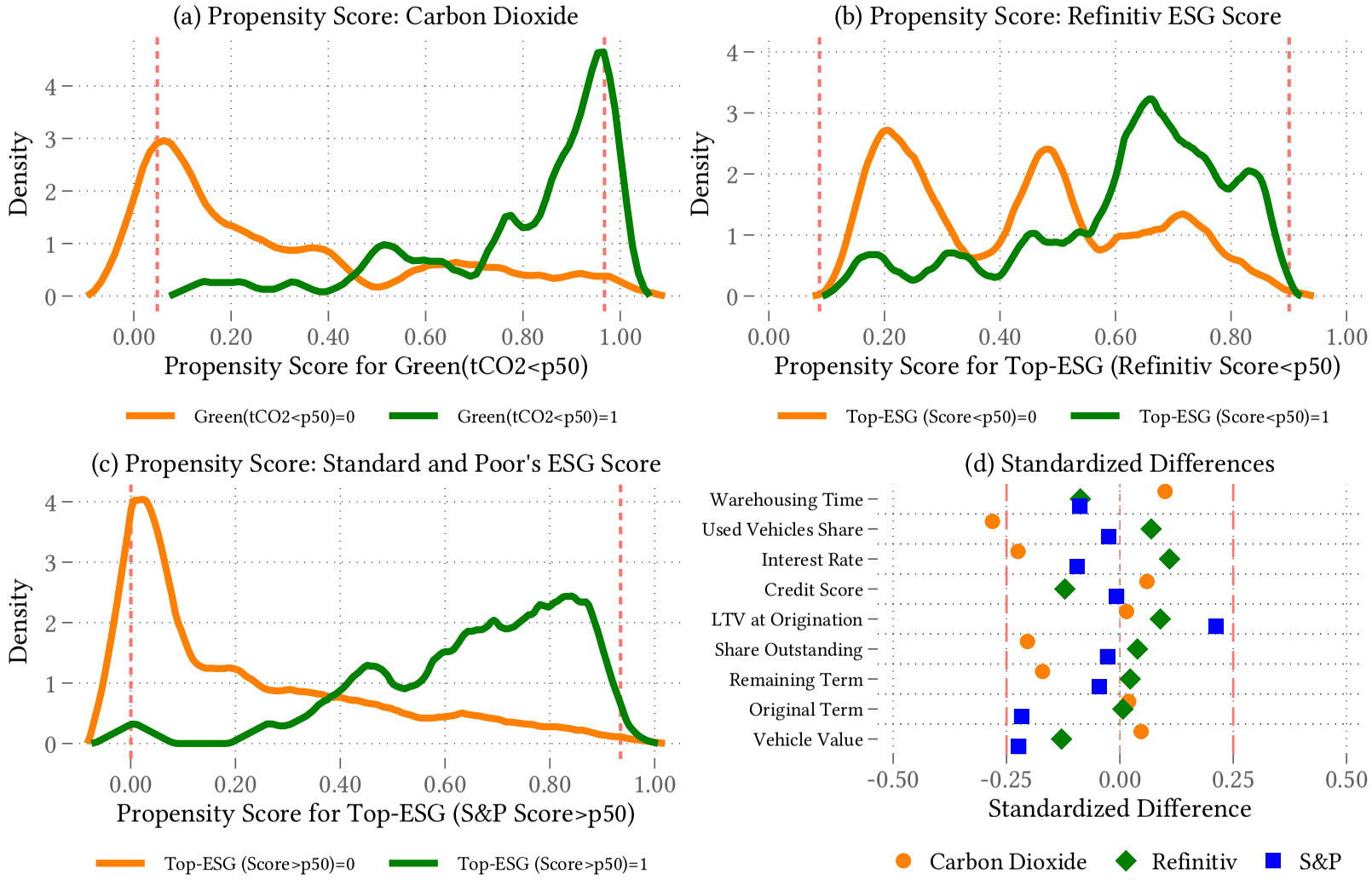



**Figure A2:** Propensity Score Matching: Diagnostics

## A.5 Double-selection Lasso Estimator

**Table A14:** Estimates using Double-selection Lasso Estimator of Belloni et al. (2014)

| | (1) Issuance Spread | (2) Issuance Spread | (3) Issuance Spread | (4) Issuance Spread | (5) Issuance Spread | (6) Issuance Spread |
|---|---|---|---|---|---|---|
| Refinitiv ESG Score | -0.511*** | -0.379*** | -0.374*** | | | |
| | (0.0672) | (0.0879) | (0.0817) | | | |
| S&P ESG Score | | | | -0.168*** | -0.163*** | -0.149*** |
| | | | | (0.0356) | (0.0424) | (0.0430) |
| Financed tCO2 per USD | -0.0729 | -0.128 | -0.102 | -0.208+ | -0.119 | -0.194* |
| | (0.113) | (0.115) | (0.0855) | (0.111) | (0.114) | (0.0982) |
| Time, Subprime, APS, Tranche FE | Yes | Yes | Yes | Yes | Yes | Yes |
| No. of potential controls | 38 | 290 | 858 | 38 | 290 | 858 |
| No. of selected controls | 11 | 15 | 15 | 11 | 17 | 15 |

Standard error clustered at year-month. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Appendix Table A14 shows that one obtains qualitatively and quantitatively similar results to the main results when using the double-selection lasso estimator of Belloni et al. (2014).

The list of potential control variables for the lasso algorithm is the following: Level of VIX at issuance, standard deviation of VIX in the 30 days before issuance, inflation expectations (5-Year breakeven inflation rate) at issuance, 6 month and 12 month estimate of the treasury yield curve from Filipović et al. (2022), attachment point, weighted average life of tranche, issuance size of tranche, total issuance size, 30d+ delinquency rate, difference to assumed prepayment speed, average share of used cars, average interest rate of loans, average warehousing time, 25th percentile of warehousing time, 75th percentile of warehousing time, average credit score of borrowers, 25th percentile of credit score of borrowers, 75th percentile of credit score of borrowers, average loan-to-value ratio at issuance, 25th percentile of loan-to-value ratio at issuance, 75th percentile of loan-to-value ratio at issuance, average % of principal outstanding at time of securitization, 25th percentile of % of principal outstanding at time of securitization, 75th percentile of % of principal outstanding at time of securitization, average remaining term, 25th percentile of remaining term, 75th percentile of remaining term, average original term, 25th percentile of original term, 75th percentile of original term, average vehicle value at origination, 25th percentile of value at origination, 75th percentile of value at origination, captive FE, US issuer FE, as well as interaction term of these variables. I require the following fixed effects to be present in each (Lasso) regression: assumed absolute prepayment speed, year-month, and subprime fixed effects.

## A.6 Pass-through elasticity

I estimate the pass-through elasticity of ABS spreads to consumer interest rates using specifications of the following form:

$$\log\left(\text{Consumer Loan Interest Rate}\right)_{i,o,t} = \beta \times \widehat{\text{ABS Spread}}_{o,t-1} + \textit{HDFE} + \mathbf{X}_i'\zeta + \varepsilon_{i,o,t}$$

The coefficient of interest, $\beta$, measures the pass-through elasticity of ABS spreads to consumer loan interest rates in percent. The relationship between ABS spreads and consumer interest rates is obviously endogenous and determined by equilibrium conditions which connect the two markets through a financial intermediary. To address this endogeneity, I employ an instrumental variable approach using exogenous shifters of funding cost for auto loan lenders. I use two different instrument as proxies for shifts in funding cost of auto loan lenders: (i) ICE BofA US Corporate Index Option-Adjusted Spread and (ii) the leave-one-out mean of auto ABS spreads issued in the same month (excluding the originator itself). The idea behind these two instruments is to exploit common variation in funding markets which correlate with the actual funding cost of individual lenders.[31]

The specifications include a set of high-dimensional fixed effects (*HDFE*) that ensures that the pass-through elasticity is identified by loans with the same characteristics made either *across issuers but within time* or *across time but within issuer*. To be specific, the *within-time across issuer* specifications include (origination month $\times$ *HDFE Set*) fixed effects. The *within-issuer across time* specifications include (originator $\times$ *HDFE Set*) fixed effects. The *HDFE Set* is given by

$$\textit{HDFE Set}: \left\{ \begin{array}{l} \text{borrower state} \times \text{vehicle type} \times \text{vehicle used} \times \text{loan term quartile} \\ \times \text{LTV quartile} \times \text{warehousing quartile} \times \text{credit score bin} \end{array} \right\}$$

where credit score bins have a width of 50. The groups defined by *within-time across issuers* and *within-issuer across time* FE have on average 24 and 497 observations, respectively. The specifications further include a vector of controls at the loan level, $\mathbf{X}_i$, that linearly control for log-transformation of maturity-matched estimates of the real yield curve, LTV ratio, payment-to-income ratio, loan term, warehousing time, vehicle value, and vehicle age. The standard errors are double clustered at collateral pool and origination month.

[31] See Berry and Haile (2021) who write "Noisy measures of a producer's actual cost shifters can also serve as instruments. For example, the average wage level in a producer's labor market may not perfectly track the producers' labor costs but is nonetheless likely to be highly correlated with those costs. Thus, such wage measures can serve as instruments as long as they are uncorrelated with demand shocks conditional on the exogenous variables [...]"

**Table A15:** Instrumental Variable Estimates of the Pass-Through Elasticity

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | within time across issuers | | | | within issuer across time | | | |
| | Dependent variable: Log(Consumer Loan Rate) | | | | | | | |
| | OLS | | IV | | OLS | | IV | |
| Spread t-1 | 0.469* | 0.424* | 0.0819 | -0.0288 | -0.00294 | -0.0199 | -0.315* | 0.102+ |
| | (0.217) | (0.172) | (0.279) | (0.307) | (0.0514) | (0.0480) | (0.127) | (0.0558) |
| Spread t-1 × Captive | 1.651*** | 1.509*** | 0.979** | 0.872** | 1.194*** | 1.261*** | 0.962*** | 0.702*** |
| | (0.263) | (0.237) | (0.313) | (0.290) | (0.186) | (0.182) | (0.272) | (0.121) |
| Total effect for Captive | 2.120*** | 1.933*** | 1.061* | 0.843* | 1.192*** | 1.241*** | 0.647* | 0.803*** |
| | (0.266) | (0.246) | (0.403) | (0.406) | (0.185) | (0.180) | (0.281) | (0.128) |
| Origin. Month × HDFE Set FE | Yes | Yes | Yes | Yes | | | | |
| Originator × HDFE Set FE | | | | | Yes | Yes | Yes | Yes |
| Year × State FE | | | | | Yes | Yes | Yes | Yes |
| Linear Controls | | Yes | Yes | Yes | | Yes | Yes | Yes |
| Instrument | | | BofA | Others | | | BofA | Others |
| Kleibergen-Paap F-stat. | | | 7.30 | 5.40 | | | 10.89 | 111.87 |
| Adj. $R^2$ | 0.916 | 0.924 | 0.107 | 0.152 | 0.896 | 0.905 | 0.093 | 0.135 |
| Sample | 7,801,359 | 7,758,215 | 7,758,215 | 5,804,920 | 9,017,159 | 8,966,592 | 8,966,578 | 6,723,651 |

Notes: Standard errors in parentheses double clustered at collateral pool and origination month. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A15 shows the estimates of the pass-through elasticity of auto ABS spreads to consumer interest rates. Several things are noteworthy. First, OLS estimates show an upward bias relative to IV estimates. Second, estimates for captive lenders are larger than estimates for non-captive lenders. In IV specifications, the pass-through elasticity is essentially zero for non-captive lenders but large and statistically significant for captive lenders. The IV specifications of Columns (3) and (4) using within time across issuer variation show that a 1 basis point increase in ABS spreads results in an increase in consumer interest rates between 0.84% and 1.06% for captive auto lenders. The IV specifications of Columns (7) and (8) using within issuer across time variation show that a 1 bps increase in ABS spreads results in an increase in consumer interest rates between 0.65% and 0.80%. These estimates imply that the green basis spread of 9 bps in the auto ABS market translates into an approximately 50 bps increase in consumer interest rate.

Appendix Figure A3 shows that captive lenders increase their supply of subsidized loans when issuance spreads in the auto ABS are low (see also Benetton et al., 2021, Klee et al., 2023). A 9 bps decrease in the issuance spreads of captive lenders results in a 6% increase in the probability of interest rate subvention for captive lenders.

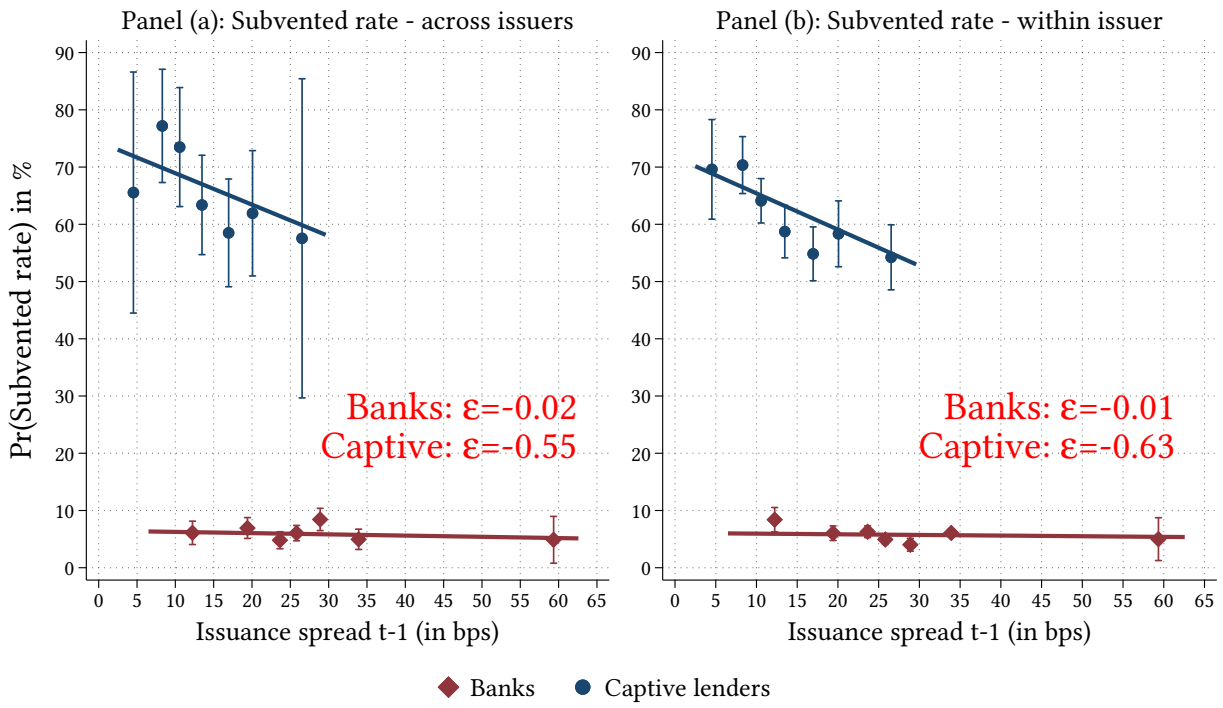



**Figure A3:** Probability of interest rate subvention depends on the issuance spreads in auto ABS market

# B Online Appendix (for online publication only)

**(a)** Santander SDRIVE 2021-4 Subprime Issue

*Pricing* $1.8bn Santander Drive Auto Receivables Trust 2021-4

Issuer: Santander Consumer USA
Lead Managers: Citi(str.), JPM, and SIS
DE&I Co-managers: AmeriVet Securities, Great Pacific Securities, Mischler Financial Group

Anticipated Capital Structure:

| CL | OFF. AMT | WAL | F/M | L.FNL | BENCH | SPRD | YLD% | CPN | PX |
|---|---|---|---|---|---|---|---|---|---|
| A-1 | $222.40 | 0.13 | F1+/P-1 | 11/15/2022 | Intl. | 7 | 0.16802 | 0.16802 | 100.00000 |
| A-2 | $543.10 | 0.64 | AAA/Aaa | 08/15/2024 | EDSF | 20 | 0.380 | 0.37 | 99.99378 |
| A-3 | $292.37 | 1.43 | AAA/Aaa | 08/15/2025 | EDSF | 15 | 0.517 | 0.51 | 99.99081 |
| B | $288.61 | 2.10 | AA/Aaa | 06/15/2026 | IntS | 30 | 0.887 | 0.88 | 99.98887 |
| C | $243.48 | 2.80 | A/Aa2 | 02/16/2027 | IntS | 47 | 1.271 | 1.26 | 99.97902 |
| D | $241.38 | 3.58 | BBB/Baa2 | 10/15/2027 | IntS | 70 | 1.685 | 1.67 | 99.98569 |
| E | $131.18 | 3.97 | NR/B2 | <<NOT OFFERED>> | | | | | |

**(b)** CarMax 2019-1 Prime Issue

$1.5bln CarMax (CARMX) 2019-1
JOINT BOOKRUNNERS : Credit Suisse (str), Barclays, Wells Fargo
CO-MANAGERS : MUFG, Scotia, SMBC, TD

| CLS | $AMT(MM) | WAL | S&P/FITCH | P.WIN | L.FNL | BNCH | SPRD | YLD% |
|---|---|---|---|---|---|---|---|---|
| A1 | 277.000 | 0.28 | A-1+/F1+ | 1-7 | 01/2020 | IntL - | 1 | 2.78007 |
| A2A | 412.000 | 1.16 | AAA/AAA | 7-22 | 07/2022 | EDSF + | 31 | 3.045 |
| A2B | 100.000 | 1.16 | AAA/AAA | 7-22 | 07/2022 | 1mL + | 31 | |
| A3 | 493.900 | 2.64 | AAA/AAA | 22-43 | 03/2024 | IntS + | 40 | 3.074 |
| A4 | 107.910 | 3.84 | AAA/AAA | 43-48 | 08/2024 | IntS + | 65 | 3.283 |
| B | 42.170 | 3.98 | AA/AA | 48-48 | 11/2024 | IntS + | 85 | 3.479 |
| C | 39.910 | 3.98 | A/A | 48-48 | 01/2025 | IntS + | 115 | 3.779 |
| D | 27.110 | 3.98 | BBB/BBB | 48-48 | 08/2025 | IntS + | 145 | 4.079 |

* Exp. Settle: 01/23/19
* First Pay Date: 02/15/19
* Px Speed: 1.30% ABS to 10% Call
* Timing: PRICED
* Format: Public/SEC
* ERISA: Yes
* Min Denoms: $5k by $1k
* B&D: Credit Suisse

**Figure B1:** Examples of Typical auto ABS Deals

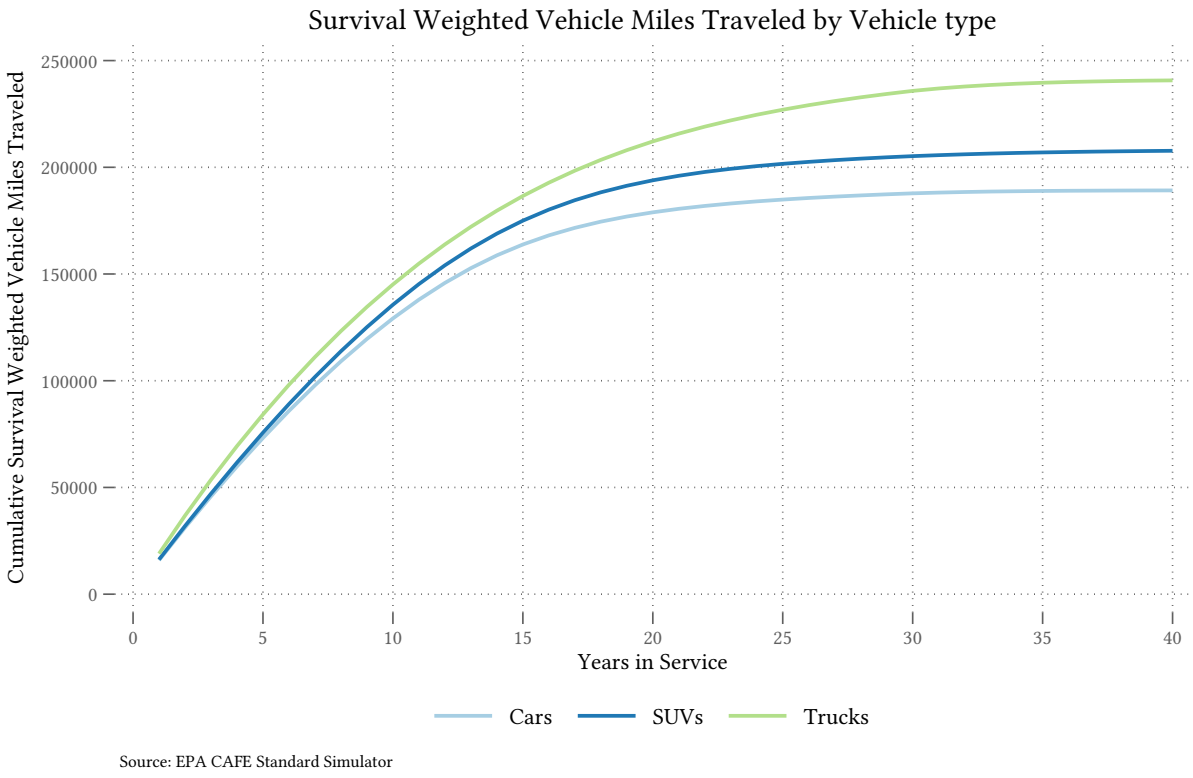



**Figure B2:** Survival-Weighted Vehicle Miles by Vehicle Type

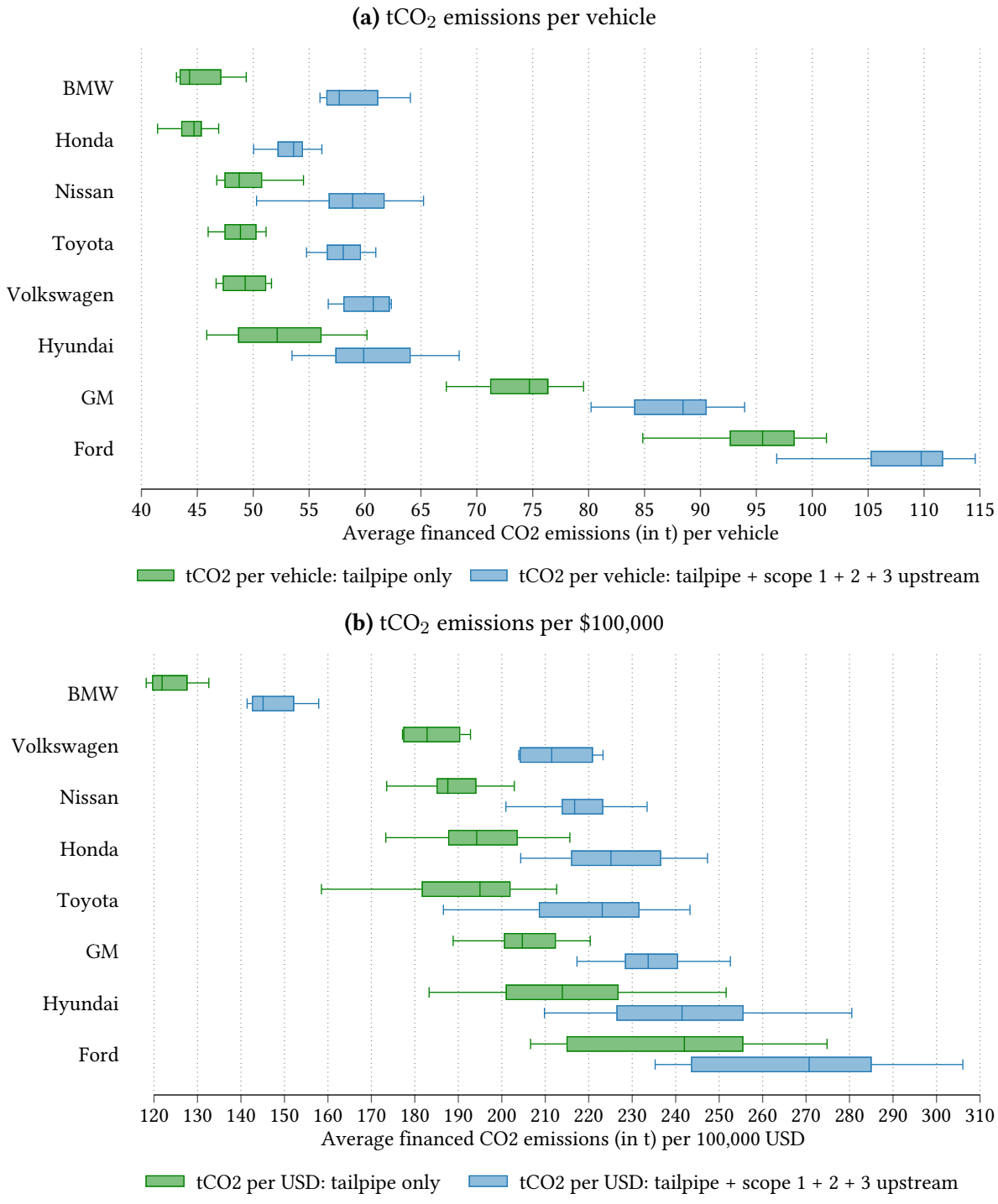



**Figure B3:** Dispersion of tailpipe and production $CO_2$ emissions across auto ABS pools of captive lenders

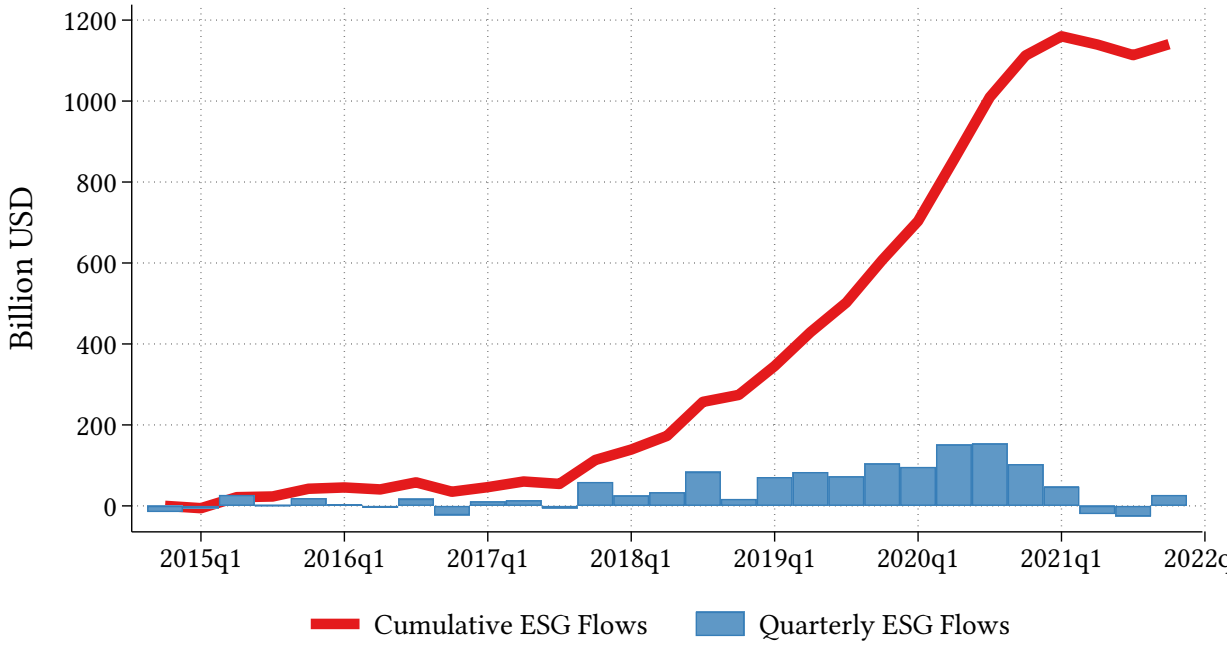



**Figure B4: Total ESG Flow (Van der Beck, 2021).** ESG flow for each 13F institution as the return-adjusted change in ESG-assets under management and then summed across all institutions. I report rolling 4-quarter averages and plot the cumulative sum of all flows since 2014.

**Table B1:** ESG Score Balance across Green ($tCO_2$ /vehicle<p50) and Brown ($tCO_2$ /vehicle>p50)

| | (1) Brown (tCO2/vehicle>p50) | | (2) Green (tCO2/vehicle<=p50) | | (1)-(2) Pairwise t-test | |
|---|---|---|---|---|---|---|
| Variable | N/Clusters | Mean/(SE) | N/Clusters | Mean/(SE) | N/Clusters | Mean difference |
| Refinitiv ESG Score | 119 | 0.783 | 124 | 0.688 | 243 | 0.095 |
| | 13 | (0.045) | 11 | (0.069) | 17 | |
| Refinitiv E Score | 119 | 0.722 | 124 | 0.659 | 243 | 0.063 |
| | 13 | (0.096) | 11 | (0.127) | 17 | |
| Refinitiv S Score | 119 | 0.753 | 124 | 0.692 | 243 | 0.061 |
| | 13 | (0.050) | 11 | (0.056) | 17 | |
| Refinitiv G Score | 119 | 0.816 | 124 | 0.676 | 243 | 0.140 |
| | 13 | (0.051) | 11 | (0.066) | 17 | |
| S&P ESG Score | 119 | 0.614 | 124 | 0.552 | 243 | 0.062 |
| | 13 | (0.082) | 11 | (0.101) | 17 | |
| S&P E Score | 119 | 0.645 | 124 | 0.582 | 243 | 0.063 |
| | 13 | (0.100) | 11 | (0.123) | 17 | |
| S&P S Score | 119 | 0.590 | 124 | 0.542 | 243 | 0.048 |
| | 13 | (0.096) | 11 | (0.112) | 17 | |
| S&P G Score | 119 | 0.615 | 124 | 0.530 | 243 | 0.084 |
| | 13 | (0.068) | 11 | (0.083) | 17 | |
| Sustainalytics ESG Score | 64 | 0.604 | 58 | 0.589 | 122 | 0.015 |
| | 11 | (0.032) | 10 | (0.030) | 16 | |
| Sustainalytics E Score | 64 | 0.567 | 58 | 0.615 | 122 | -0.048* |
| | 11 | (0.054) | 10 | (0.045) | 16 | |
| Sustainalytics S Score | 64 | 0.650 | 58 | 0.557 | 122 | 0.093 |
| | 11 | (0.028) | 10 | (0.043) | 16 | |
| Sustainalytics G Score | 64 | 0.608 | 58 | 0.587 | 122 | 0.021 |
| | 11 | (0.036) | 10 | (0.022) | 16 | |

Notes: Pairwise t-tests adjust for industry fixed effects. Standard errors are clustered at issuer-level. * p<0.05, ** p<0.01, *** p<0.001.

**Table B2:** Correlation of ESG scores and measures of environmental impact in N-PORT holdings

| | Refinitiv ESG Score | S&P ESG Score | Financed tCO2/car | Financed tCO2/USD | Avg. MPG | Truck % | GHG Rating |
|---|---|---|---|---|---|---|---|
| Refinitiv ESG Score | 1.00 | | | | | | |
| S&P ESG Score | 0.85 | 1.00 | | | | | |
| Fin. tCO2/car | 0.50 | 0.41 | 1.00 | | | | |
| Fin. tCO2/USD | 0.36 | 0.34 | 0.54 | 1.00 | | | |
| Avg. MPG | 0.32 | 0.25 | 0.83 | 0.42 | 1.00 | | |
| Truck % | 0.33 | 0.24 | 0.83 | 0.33 | 0.86 | 1.00 | |
| GHG Rating | 0.27 | 0.15 | 0.75 | 0.19 | 0.86 | 0.85 | 1.00 |

MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse.